

Implementation of Communication and Group Formation in the Agent Based Modelling of Forced Migrants: Mali Case Study

[Anonymised review copy]

Abstract: Effective forced migration forecasting is pivotal for planning humanitarian responses. Traditional models like Flee (Groen 2016), though widely utilised, depend heavily on complete refugee registration data from the United Nations High Commissioner for Refugees (UNHCR), limiting proactive capabilities and under-utilising insights from migrant behaviour. This research introduces the Grouping and Rumours of Asylum-Seeking Persons (GRASP) model, which incorporates agent grouping and communication mechanisms to address these limitations. The model incorporates both physical and network-based path distances between nodes and uniquely integrates dynamic border status changes over time, capturing real-world movement constraints and resistance across international borders. Tested against the 2012 Mali crisis using data from the Armed Conflict Location and Event Data (ACLED 2024) database, GRASP independently predicted refugee numbers in camps with 77.85% average daily precision. Furthermore, GRASP's ability to detect notable population changes aids in the timely deployment of humanitarian aid. The model also classifies displaced populations with high accuracy. A key innovation of GRASP is its modular architecture, enabling users to adapt core mechanisms such as grouping, communication, and decision logic with minimal effort for rapid application to new contexts. Our findings indicate that integrating empirical grouping improves precision by 48.8% and the communication mechanics implementation reduces relative difference by 61.26% on average. This suggests a path forward for more sophisticated and effective migration forecasting tools. GRASP can be applied to any context with publicly available data within a day, with further granularity achievable within 2–3 days, such as refining utility preferences or increasing the number of nodes. This study highlights the critical role of advanced agent-based modelling in improving humanitarian response strategies.

Keywords: Agent Based Modelling, Forced Migration, Simulation, Humanitarian

● Introduction

- 1.1 The United Nations High Commissioner for Refugees (UNHCR) defines forced migration as situations in which individuals or communities are compelled to leave their homes due to factors such as armed conflict, generalised violence, human rights abuses, natural or man-made disasters, or development projects (United Nations High Commissioner for Refugees (UNHCR) 2008). Since the end of the Second World War, the number of forcibly displaced people has risen, reaching 123.2 million by the end of 2024. Among these, nearly 42.7 million were refugees¹ and 73.5 million were Internally Displaced People (IDP)² (United Nations High Commissioner for Refugees (UNHCR) 2025a). Their numbers are expected to continue increasing due to growing inequality, decreasing global stability, and the impacts of climate change (Myers 2002). By 2030, 46% of the global population is anticipated to reside in conflict-affected areas (Network 2017).
- 1.2 These individuals often forced from their homes and cities, flee from a myriad of humanitarian disasters, including local conflicts, human rights violations, serious disturbances of public order, persecution, and violence (UNHCR 2022a). While scholars have historically identified violence as a key driver of displacement (Moore & Shellman 2007; Schmeidl 1997; Shellman & Stewart 2007), more recent analyses highlight the growing role of climate-related and environmental factors (Internal Displacement Monitoring Centre (IDMC) 2024).

¹Defined in this research paper as forced migrants who leave their country of origin and do not return.

²Defined in this research paper as forced migrants who remain in their country of origin.

- 1.3 The dynamics of forced migration are uncertain and highly complex, they are considered the most un-predictable drivers of population change (National Research Council 2000). Mathematical modelling can help describe observed patterns and simulate possible outcomes of these migration dynamics, particularly when direct experimentation with policy implementations is impractical (Borshchev & Filippov 2004). Broadly speaking, such modelling is used to either investigate causality, extrapolate information, or highlight a statistical advantage in an argument/hypothesis (Bijak 2022). However, several studies highlight limitations in current migration modelling approaches, including weak theoretical foundations, methodological biases, and insufficient treatment of complex driver interactions (Massey et al. 1993; Davenport et al. 2003; Van Hear et al. 2018; Cheong 2023).
- 1.4 Forecasting models for migration have been developed since the first attempts to formalise migration analysis (Ravenstein 1885). With the advancement in conceptualising population growth and decline mechanisms, improvement of modelling techniques, and rapid development of computational power, demographic forecasting has evolved to increasingly fine levels of granularity (Willekens 1990; ?).
- 1.5 Cheong (2023) argues that progress in the field of forced migration forecasting has been insufficient, partly due to a biased perception of refugees from researchers in host countries. These researchers often overlook the contributions and experiences of forced migrants and their countries of origin, employ methodological nationalism³, and focus predominantly on 'policy-related research' while maintaining a traditional division between quantitative and qualitative approaches. These limitations can constrain the development of more comprehensive and representative migration models.
- 1.6 The theoretical understanding of the drivers behind migration has long been recognised as limited, despite their significant impact across many countries (Massey et al. 1993). More recent literature continues to highlight challenges in capturing the complexity and interaction of migration drivers. Historical models of migration have also been criticised for sample selection bias and insufficient attention to 'pull' factors (drivers in potential destination countries that attract forced migrants) (Davenport et al. 2003).
- 1.7 Existing models often overlook the role of group dynamics and communication among migrant populations, with many migration models focusing primarily on individual-level decision-making or aggregate flows rather than interactions between agents. These elements are essential for understanding certain migration patterns, particularly when people travel together and share information to guide decisions about potential destinations, as highlighted in studies of migration networks and information flows. Without accounting for such interactions, these types of models struggle to capture the realities of these forms of forced migration, leaving a gap in the literature. At the same time, it is important to recognise that group-based movement is not universal. It is most commonly linked to acute-onset events, such as sudden outbreaks of violence or climate-related disasters. In other contexts, especially in some waves of displacement, solo migration is more prevalent. For instance, during the early stages of the 2015 refugee crisis in Lesbos, Greece, the first arrivals were often single men traveling alone, while family groups became more prominent in later waves as demographics shifted (United Nations High Commissioner for Refugees (UNHCR) 2015; Crawley et al. 2016).
- 1.8 The aim of this research is to enhance the accuracy of an ABM for forecasting the movement of forced migrants by integrating group dynamics and social networks. Using the case study of Mali in 2012, the research seeks to demonstrate that a more granular understanding of the forced migrant experience leads to improved predictive outcomes, particularly in modelling camp population distributions, destination selection, and the temporal dynamics of displacement, through the incorporation of grouping mechanisms and communication processes, compared to baseline models such as Flee.
- 1.9 To aid in both the reconstruction of GRASP and to provide a deeper understanding of the inner mechanisms and assumptions, this paper includes a dedicated appendix for Overview, Design concepts, Details (ODD) and an appendix on Supplementary Notes. For brevity, the Supplementary Notes will be used to explain concepts and assumptions referenced in the main text in higher detail.

● Summary of literature review

Motivation

- 2.1 Forecasting forced migration is essential for estimating displaced populations, identifying likely destinations, and supporting humanitarian planning and resource allocation (Edwards 2008; Frydenlund 2021; Suleimenova

³Methodological nationalism assumes that nations are natural entities, often taking national discourses for granted without analytical focus beyond national boundaries (?).

et al. 2022). Traditional analytical and econometric migration models have contributed substantially to understanding long-term migration processes but often struggle to represent sudden conflict-induced displacement and heterogeneous individual decision-making (Bijak et al. 2019; de Haas 2021). Consequently, Agent-Based Models (ABMs) have become an increasingly important methodology within forced migration research because they explicitly represent individual agents, their interactions, and the emergence of system-level migration patterns (Borshchev & Filippov 2004; Macal & North 2010). These characteristics make ABMs well suited to evaluating humanitarian scenarios and policy interventions under rapidly evolving crises (Frydenlund 2021).

- 2.2** Among forced migration ABMs, the Flee framework has become one of the most influential models, providing a flexible network-based simulation capable of forecasting refugee movements using publicly available humanitarian datasets (Groen 2016; Suleimenova et al. 2017a, 2022).

Flee Model

- 2.3** The Flee model, developed by Groen (2016) and integrated into the ITFLOWS project (ITFLOWS 2022), predicts over 75% of refugee destinations after the first 12 days of displacement using UNHCR and ACLED data (Suleimenova et al. 2022). However, it assumes individuals travel alone, neglecting group dynamics and communication, and relies heavily on registered refugee data, excluding unregistered migrants and IDPs (Suleimenova et al. 2017b). The model, insofar as it does not consider the interaction between agents and groups, is missing a key component of the real-world behaviour of agents, whose choices are affected by the choices of other agents.

Metafora

- 2.4** Metafora (Cheong 2023) is an agent-based model of forced migration where migrants, helpers, and officials interact across a network of nodes. Migrants are defined by resources, information, traits, and group size, while helpers include guides, smugglers, personal ties, governments, and NGOs. The model tracks expenditures and node factors such as attraction, conflict, employment, and border rules. Movement follows cognitive threshold logic (Granovetter 1978) and accounts for strategic, tactical, and operational relocation. Though ethical assumptions are addressed, the model omits dynamic grouping, non-smuggler communication, and applies only at the international scale.

Limitations of existing models

- 2.5** Although recent ABMs have substantially advanced forced migration forecasting, several limitations remain. Existing models often simplify behavioural processes, rely heavily on registered refugee data, or place limited emphasis on interactions between migrants. These limitations motivate the incorporation of empirically informed grouping and communication mechanisms within GRASP (Cheong 2023; Suleimenova et al. 2017a; Frydenlund 2021).
- 2.6** Group formation in forced migration is shaped by trade-offs between speed, safety, and trust in shared route information (Collins & Frydenlund 2016). Forced migrants often rely on networks of trust built from earlier migrants' experiences, influencing both routes and timing (UNHCR 2017; Hinsch & Bijak 2019; Cheong 2023). Models such as Routes & Rumours (Bijak & Hinsch 2021) demonstrate how information exchange and perceived risks can drive collective movements, but their validity depends on grounding assumptions in empirical evidence (Frydenlund 2021).
- 2.7** Communication, via face-to-face contact, mobile phones, or social media, can dramatically alter migration patterns as earlier arrivals relay route and destination information to those still in transit (Alencar 2018; Gillespie et al. 2018; Dekker et al. 2018). In some corridors, such as Venezuelan migration to Colombia and Peru, or Mediterranean and Afghan–Turkish routes, smugglers act as dominant intermediaries, shaping or constraining movement decisions (Mixed Migration Centre (MMC) 2021; Achilli 2018; Sanchez 2018). These dynamics justify modelling both grouping and communication in GRASP, while also highlighting contexts where communication is mediated (or overridden) by actors beyond migrants' control (see Section 6).

Methodology

Model overview

- 3.1** GRASP (Grouping and Rumours of Asylum Seeking Persons) is an agent-based model of forced displacement. Each agent represents a person moving across settlements, camps, borders, and roads, making daily choices that balance safety, distance, capacity, and received information. The model can run on a subsample of the population while retaining individual-level behaviour, with aggregate outputs validated against observations.
- 3.2** Key mechanisms include dynamic grouping, communication of route and risk information, and stochastic decision-making based on conflict proximity, travel costs, accessibility, and information. Its modular design allows testing the impact of each component, and it extends Flee (Groen 2016) by incorporating individual-level grouping, information exchange, and richer choice logic to improve predictions of population flows.

Mali

- 3.3** At the heart of the Central Sahel, Mali faces multifaceted challenges including inadequate security, conflict, and climatic shocks. According to United Nations High Commissioner for Refugees (UNHCR) (2025b) statistics, in 2012 (the year of the case study) Mali hosted approximately 14,000 foreign refugees while about 150,000 Malian nationals were displaced abroad as refugees (UNHCR 2024). By 2024, these figures had risen to around 136,000 foreign refugees hosted in Mali and 354,000 Malian refugees residing in other countries (UNHCR 2024). Beyond displacement, Mali grapples with severe environmental hazards such as rising temperatures, projected to increase by 2.5°C by 2080 (Wachiaya 2021), as well as risks of deforestation, desertification, and soil erosion that compromise water availability. Human trafficking also remains prevalent along migration routes, with digital technologies playing a dual role as both a risk factor and a resource for safety information (UNHCR 2022b). Regionally, ongoing conflicts in the Central Sahel have led to more than 410,000 refugees and approximately 2.1 million internally displaced persons (Cheshirkov 2022).

Data handling

- 3.4** The raw data used in this model are publicly available and builds on that used in the Flee model (Groen 2016), with additional data for improved granularity in the model implementations:
- Armed Conflict Location and Event Data (ACLED) (ACLED 2024)
 - UNHCR for: (UNHCR 2024)
 - Refugee data
 - Family size distribution (Sudan)
 - Group type distribution (Moldova)
 - Communication source distribution (Myanmar)
 - City Population (City Population 2022)
 - City population distribution
 - Demographic data
 - Wealth distributions (The World Bank 2022; World Inequality Database 2022; Oxfam in West Africa 2023)

The ACLED (2024) data offer details on conflicts in Mali from 06/01/1997 to the present, including date, type of disorder/event, actors involved, civilian targeting, region, geographic coordinates, and fatalities.

UNHCR (2024) reports provide information on camp locations and populations, border openings to Niger and Burkina Faso on 21st February 2012, and the closure of the Fassala camp on 19th March 2012. To compare with Flee, the model created uses data from 1st January 2012 to 1st January 2013, with temporal points linearly interpolated. Initial population values for the model on 1st January 2012 were extrapolated from UNHCR data.

The camp locations and population data are drawn from the closest camps in bordering countries, which were also the direct examples used in the original Flee model (Groen 2016), ensuring consistency with prior modelling approaches for the Mali case study.

City populations for 2012 were linearly interpolated from the 2009 and 2022 censuses, assuming constant gender and age distributions from the 2022 census. Wealth distribution was estimated using data from The World

Bank (2022), World Inequality Database (2022), and anecdotal evidence from Oxfam in West Africa (2023) about the capital of those in the upper 0.1% of the country's wealth distribution.

Micro-data were extracted from UNHCR case studies on Sudan, Moldova, and Myanmar by Cheong (2023). Household size distributions, though not directly related to Mali, are comparable to Sudan due to similar mean household sizes (The DHS Program 2020; ESRI 2023). Moldova and Myanmar were the only available sources for data on forced migrant travel groups and their sources of information. For the purpose of this paper, which focuses on the methodological framework rather than the exact replication of migrant distributions in Mali, we assume that migrants in Mali behave similarly to those observed in the Moldova and Myanmar case studies.

Flee benchmark

- 3.5** Flee's simplicity and quick instance construction are notable strengths. However, GRASP aims to outperform Flee by enhancing accuracy and precision in predicting refugee numbers (how many flee) and reducing relative error in the camp population distribution (where they flee to). Flee tends to underestimate the number of refugees (Groen 2018) and relies heavily on UNHCR refugee registration counts. Thus, the new model seeks to improve upon Flee by classifying refugees in camps with higher precision.
- 3.6** To improve accuracy and reduce dependency on external counts, GRASP can be developed to respond solely to ACLED (2024) data on conflicts. This would enable more immediate humanitarian responses and gain deeper insight on the micro-processes behind a forced migrant initially fleeing. The developing model also aims to forecast the number of forced migrants, IDPs, returnees, and refugees in specific destinations.

Validation

- 3.7** The output data of both Flee and GRASP will be compared to the UNHCR (UNHCR 2024) camp data, with GRASP utilising more UNHCR camps in the model output. The camps, in which UNHCR (2024) data are available, featured in both models is shown in Table 1 and Fig. 1.

Camp Name	Country	Flee	GRASP
Bobo	Burkina Faso	Included	Included
Goudoubo	Burkina Faso	Not included	Included
Mentao	Burkina Faso	Included	Included
Ouagadougou	Burkina Faso	Not included	Included
Fassala	Mauritania	Not included	Included until closure
Mbera	Mauritania	Included	Included
Abala	Niger	Included	Included
Mangaize	Niger	Included	Included
Niamey	Niger	Included	Included
Tabareybarey	Niger	Included	Included

Table 1: UNHCR camps (UNHCR 2024; Groen 2016)

- 3.8** GRASP, unlike Flee, will estimate the destination country distribution of refugees fleeing to foreign settlements as well as camps, validated using UNHCR (UNHCR 2024) data. Similar to Flee, the relative errors of camp populations (scaled with $N_{data,all}/N_{sim,all}$) will be calculated using Equation 1.

3.9

$$\text{Relative Error} = \left| \frac{\text{Empirical} - \text{Output}}{\text{Empirical}} \right| \quad (1)$$

- 3.10** After validating GRASP's performance, it will be crucial to run alternate versions bypassing modules like grouping, communication, and the decision choice model (see Supplementary note 3). This helps assess the model's dependency on these features for accuracy, forming an argument for their inclusion in future ABMs.
- 3.11** Estimating the number of forced migrants can be viewed as a form of binary classification (see Supplementary note 5). For this reason, in our analysis, we will focus on precision and accuracy. Precision is the ratio of correctly classified positive outcomes to total predicted positive outcomes, while accuracy is the total number of correctly predicted outcomes (Sirsat 2019).

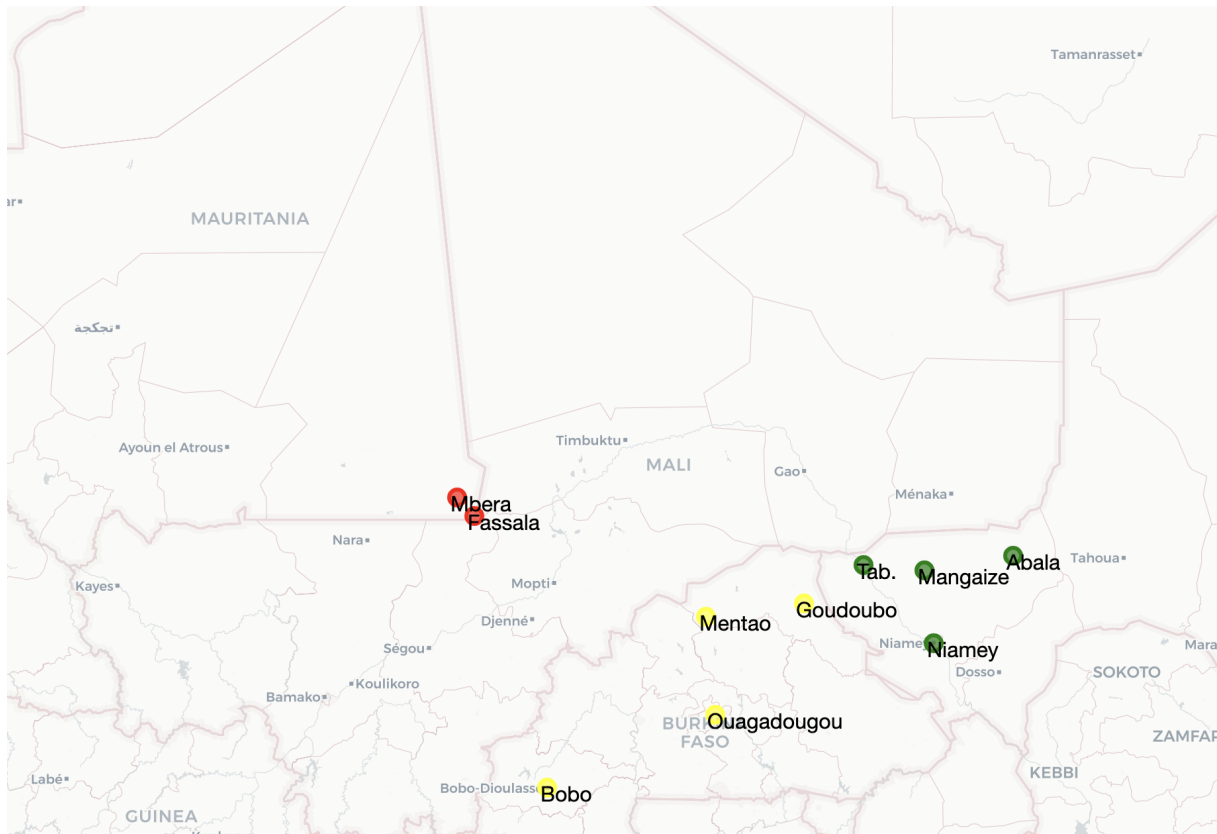


Figure 1: Camp locations. Red camps are in Mauritania; yellow, Burkina Faso; and green, Niger.

- 3.12** Unless otherwise stated, summary precision values reported in the paper refer to the arithmetic mean of the daily precision values across the evaluation period. Where pooled precision is reported, true positives and false positives are first aggregated across all time steps before calculating a single precision value.
- 3.13** The total refugee population in camps from the simulation can be compared to UNHCR data, considering the binary value of being in a camp or not. This logic applies to both the number of refugees in camps, total number of refugees, and number of IDPs.
- 3.14** Unless otherwise stated, all quantitative performance metrics reported in this paper are calculated only at empirical UNHCR observation dates. Linearly interpolated values are displayed in selected figures solely to aid visual interpretation of temporal trends and are not used in the calculation of any reported accuracy, precision, or relative difference metrics.
- 3.15** In all figures comparing model outputs with UNHCR data, empirical observation dates are indicated by vertical dash-dotted lines, allowing the distinction between measured observations and interpolated values used for visualisation.
- 3.16** While these methods determine case accuracy, empirical validation of micro-processes is necessary for creating a flexible, unbiased model. As Cheong (2023) argues, this involves disregarding agent logic and geospatial cell states. Trace validation, where Subject Matter Experts (SMEs) analyse forced migrants' movements (Gore et al. 2017), can also be implemented in future stages. SMEs were not used in this case study due to limited access and to maintain consistency with the control setup based on the Flee model.

● Agent-based model of Mali

Environment

- 4.1** A network-based environment was chosen over a spatial grid to enhance performance with a large number of agents. It consists of 92 nodes, including the top 40 populous cities in Mali, all public Malian airports, UN-HCR camps, and foreign towns on the Malian border. 405 edges were manually selected, with distances calculated using Dijkstra's shortest path algorithm. For individual links, the Haversine equation was used to calculate straight-line distances on a spherical plane (see Supplementary note 7).
- 4.2** Key temporal aspects of the network include conflict zones and border statuses. When a conflict is identified from the ACLED (2024) database, the location becomes a conflict zone (or the nearest available location). The *has_conflict* attribute becomes true for 10 days, and the *fatalities* attribute increases by the recorded number of fatalities. Border statuses can be updated in real-time, with Niger and Burkina Faso only opening on 21st February. Camp closures, such as Fassala closing on 19th March 2012, are also updated. In such cases, agents may travel through the zone but not rest or accept the camp as their final destination.

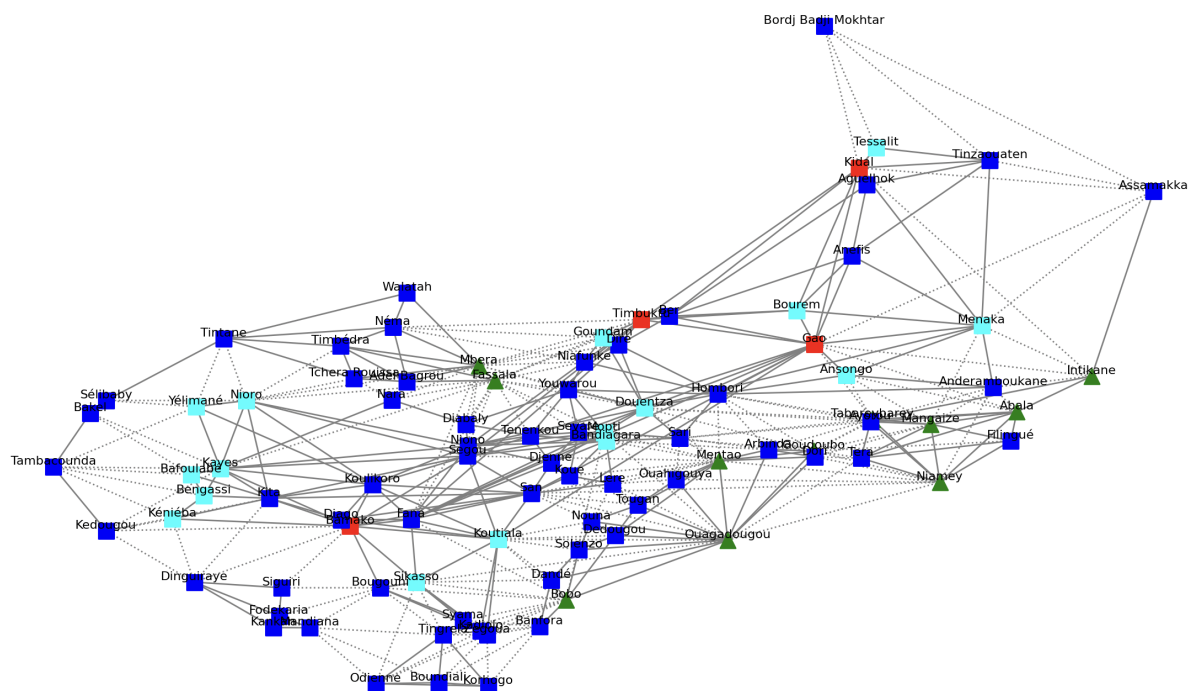


Figure 2: Network on 1st January 2013. Rectangular nodes represent cities. Triangular nodes represent camps. Cyan indicates an airport present. Red indicates a conflict within the past 10 days. Dotted link indicates the crossing of a boarder.

Agents

- 4.3** The age and gender distribution of Malian agents in 2022 is assumed to mirror that of 2012.
- 4.4** Fig 3 displays the 2012 Malian wealth distribution, calculated by interpolating data from The World Bank (2022), World Inequality Database (2022), and Oxfam in West Africa (2023). Individuals' capital is assumed to be 10 times their annual income. According to Oxfam in West Africa (2023), "in Burkina Faso, Ghana, Mali and Niger, the richest 0.1% hold at least twice as much wealth as the bottom 50% combined." This contextual evidence aids in estimating the upper end of the wealth distribution due to the lack of publicly available data.
- 4.5** In the model, wealth influences agents' transport speed and destination selection. Wealth is assigned using a percentile-based transformation of the form:

$$W(p) = 100000 \cdot (100 - p)^{-0.25} - 30000, \quad (2)$$

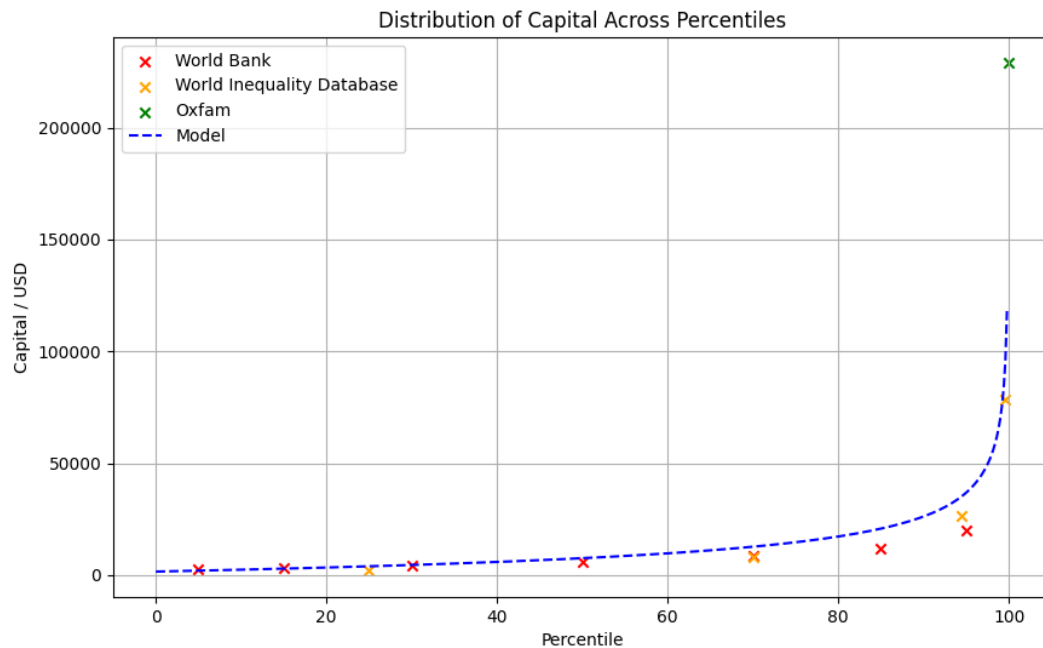


Figure 3: Estimated distribution of capital in Mali 2012 (The World Bank 2022; World Inequality Database 2022; Oxfam in West Africa 2023)

where $p \in [0, 100]$ denotes the agent's wealth percentile. Age affects both mobility and dependent classification (relevant to group formation), while gender is not currently used but remains a possible future extension.

- 4.6** Unlike Metafora (Cheong 2023), which defines grouped and individual agents, GRASP simulates individuals within groups as independent agents. This approach allows for greater heterogeneity, leadership dynamics, and the inclusion of birth and death processes for each agent within the travelling groups.

Topology

Group Formation

- 4.7** GRASP utilises the micro-data provided by UNHCR on the Moldova crisis (Cheong 2023), outlining the distribution of group types through refugee interviews. Although further validation with Malian data is needed, the data serves as a standard, challenging the Flee assumption of agents travelling alone. Within GRASP, only 30% of agents travel alone, 60% travel with their family, and 10% form strategic groups.
- 4.8** Where possible, solo travellers are aged between 16 and 65. Logic is in place to assign dependent agents to families initially and for independent agents not in a camp to seek groups to travel with. An agent in a family, even if designated as a leader, can choose to travel independently. If so, they will store their family as a contact and notify them upon reaching their destination. Using Bayesian calculations, given their associated probabilities, the probability of an agent travelling with their family, given that they are of independent age and have a family, is around 17.6%.
- 4.9** Family sizes are defined according to micro-data from Sudan⁴. Families are formed at the start of the simulation, with the possibility of increasing in size through births and decreasing through deaths. Newborn agents are immediately assigned to a family group and do not make independent migration decisions; instead, their movement is fully determined by the behaviour of the group to which they belong. While births are included primarily to support long-term simulations and group cohesion, their impact is minimal over shorter time frames. A limitation of the current model is that there is no age-based transition to independent decision-making. Deaths,

⁴Ref

however, play a more active role—especially in conflict zones—where they influence both agent survival and the dissemination of risk information to other agents.

4.10 Family leaders conduct decision-making on behalf of the group. Unlike strategic groups, these families remain together throughout their journey (excluding deaths). Independent agents have the option to travel separately from their family.

4.11 Strategic groups are formed with sizes $N \sim N(10, 4^2)^5$. Inspired by Collins & Frydenlund (2016), agents assess group utility at each node in their journey. As Collins & Frydenlund (2016) utilised a spatial grid rather than a network-based graph, only the kick-out and super group mechanisms are implemented. Initially, a Characteristic Function Game (CFG) was considered for group splitting (Rahwan et al. 2015) but it proved too computationally complex for the high number of agents. This choice had a notable impact on the model's output, with higher levels of granularity in agent sizes producing more realistic group behaviours. However, the additional complexity required was not justified for the present implementation.

4.12 Kick-out mechanism

4.13 The kick-out mechanism is determined by comparing group utilities before and after removing the slowest member.

$$V = \lambda \times u_n + (1 - \lambda) \times (N - 0.5)/N \quad (3)$$

$$V' = \lambda + (1 - \lambda) \times (N - 1.5)/N \quad (4)$$

$$u_n = \frac{u_1}{u_2} \quad (5)$$

Where Eq. 3 is the group utility, Eq. 4 is the group utility after removing the slowest member, u_1 is the movement speed of the slowest member, u_2 is the movement speed of the second slowest member, u_n is the normalised movement speed, N is the group size, and λ is the weighting (or intolerance) to which the agents prioritise speed over group size. If $V' > V$, the slowest member is removed.

4.14 Super group mechanism

4.15 The super group mechanism compares the utilities of individual groups and a combined group. If the super group has a higher utility than both individual groups, they merge.

$$V_1 = \lambda \times u_{n,1} + (1 - \lambda) \times (N_1 - 0.5)/(N_1 + N_2) \quad (6)$$

$$V_2 = \lambda \times u_{n,2} + (1 - \lambda) \times (N_2 - 0.5)/(N_1 + N_2) \quad (7)$$

$$V_{12} = \lambda \times u_{n,12} + (1 - \lambda) \times (N_1 + N_2 - 0.5)/(N_1 + N_2) \quad (8)$$

$$u_{n,1} = \min\left(\frac{u_1}{u_2}, 1\right) \quad (9)$$

$$u_{n,2} = \min\left(\frac{u_2}{u_1}, 1\right) \quad (10)$$

$$u_{n,12} = \min(u_{n,1}, u_{n,2}) \text{ or } \min\left(\frac{u_1}{u_2}, \frac{u_2}{u_1}\right) \quad (11)$$

Where Eq. 6 and Eq. 7 are the group utilities of group 1 and 2, Eq. 8 is the group utility of the super group, u_1 and u_2 are the movement speeds of group 1 and 2, $u_{n,1}$, $u_{n,2}$, and $u_{n,12}$ are the normalised moving speeds of the groups 1, 2, and super group, and N_1 and N_2 are the respective sizes of group 1 and 2.

4.16 Collins & Frydenlund (2016) found λ to be most realistic at 0.225, which is adopted in GRASP until an agent becomes stuck. In such cases, agents will attempt to increase their λ value, becoming less willing to remain in larger, slower groups in order to resume movement.

Communication

4.17 At each node, agents communicate with up to two other agents and perform the following behaviours: 1) share their long-term destination to increase familiarity with potential destinations, 2) merge their *no-go nodes*, and 3) store the agent in their contact list if they rely on contacts for destination information.

⁵This, again, is another proxy that needs to be validated with qualitative interviews.

4.18 Agents with contacts, whether from meeting other agents or parting from family, notify them when they reach a camp. Approximately 6% of agents gain information from contacts, and about 36% from family members, extrapolated from UNHCR micro-data from Myanmar (Cheong 2023).

Decision model

4.19 The destination location for each agent is continuously updated, responding to changes in conflict zones, improved information from communication, group dynamic changes, and stochastic variations in journey planning. The agent's capital bracket influences their destination: rich agents aim for the nearest five airports, middle-bracket agents aim for settlements in neighbouring countries (finding the closest settlement per country), and poor agents aim for camps⁶. This is similar to the multi-modal travel model employed by Hébert et al. (2018).

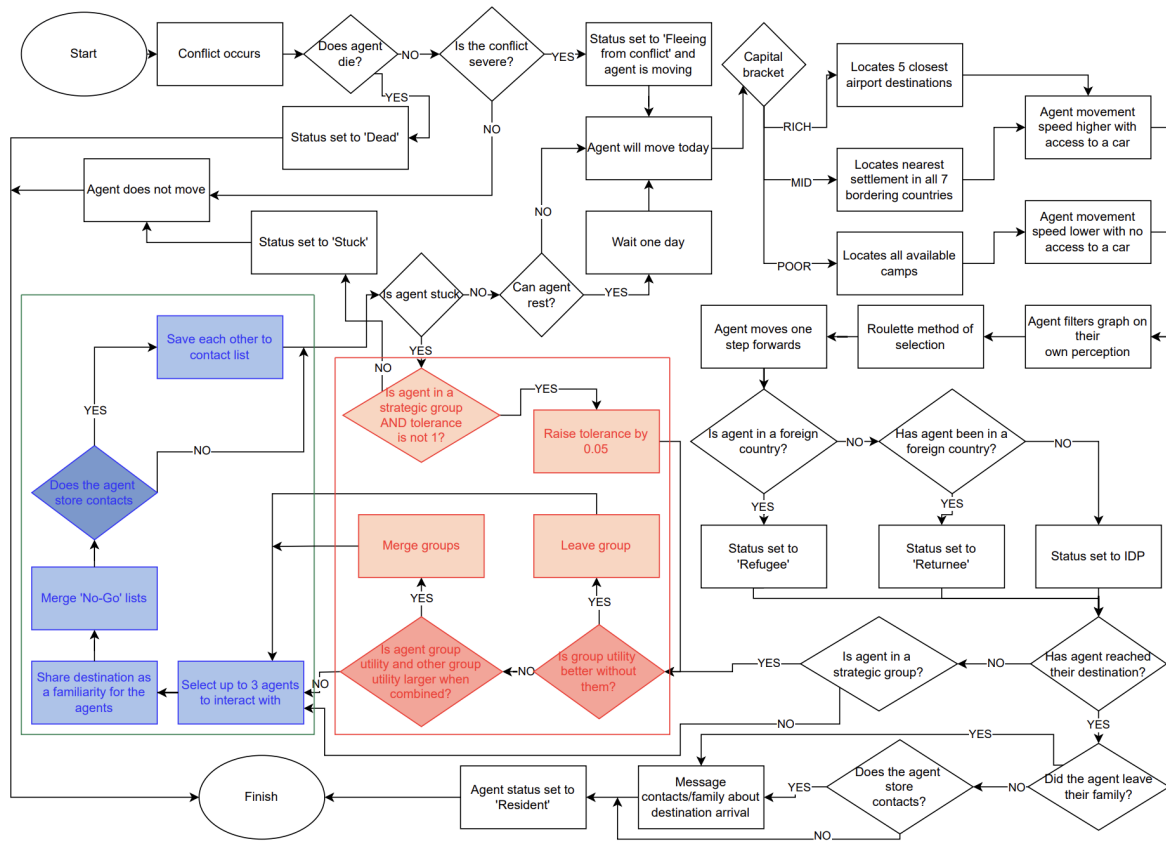


Figure 4: Agent decision logic. The red partition denotes the strategic group logic. The blue partition denotes the communication logic.

4.20 Agents review their options using the following utility equation:

$$\begin{aligned}
 V_i = & \beta_0 + \beta_{F,i}F_i - \beta_{x,i}x_i - \beta_{P,i}P_i \\
 & + \beta_{\theta,i} \cos(\theta_0 - \theta_i) \\
 & - \beta_{D,i}D_i + \beta_{N_1,i}N_{1,i} + \beta_{C,i}C_i
 \end{aligned} \tag{12}$$

Where F_i is the familiarity for alternative i (how many agents have told them about this destination), x_i is the shortest network distance to alternative i , P_i is the camp population (taken as 0 if not a camp), θ_0 is the directional bearing of the agent, θ_i is the directional bearing of the first link in the path⁷, D_i is the associated death toll of destination country, $N_{1,i}$ is the number of agents who travelled on the first link of the path (inspired by Social Network Theory (SNT) (Simon 2019)), C_i is the number of contacts who have made it to alternative

⁶Capital greater than \$100,000 USD is considered rich, and capital less than \$10,000 USD is considered poor.

⁷Bearings were implemented after recursive cycles were identified in some agents.

i , and the β values are parameters to be tuned. In principle, if real-world trajectories sampled at a daily frequency were available, these β parameters could be learned directly from data. A bias, β_0 , term was arbitrarily selected but it can be interpreted as a parameter of irrationalism, increasing the likelihood of rarer events. This is an example of an additive compensatory holistic model (Simon 1955) - one that evaluates decision-making attributes in a non-sequential manner, allowing high performance in one attribute to offset low performance in another; it acknowledges the bounded rationality of forced migrants as they navigate investments and risks under cognitive constraints when choosing destinations and routes (Bonabeau 2002; Dillon 1998).

- 4.2.1** GRASP was coded in Python 3.11 utilising packages such as OSMNX and NetworkX. It is a network-based ABM simulating daily increments and agent movements from node to node to reach a destination. During each time step, agents produce a list of potential destinations (with group decisions taken by the leader). The agent scores the destinations based on a utility function, where the score represents the probability of choosing that destination. Agents initially flee when the conflict in their location exceeds their unique fatality threshold. At each iteration, agents can either rest or move through one or more links, with a maximum daily movement speed attributed to the agent (or the slowest agent in their group). At each node, strategically grouped agents consider merging with other groups or removing the slowest member. In the same time step, each agent interacts with up to two additional agents, sharing information about common destinations and conflicts in nodes they have explored directly or indirectly as well as acquiring contacts (only applicable to 6% of population). Agents continue moving until arriving at a destination, which they leave only in the case of conflict or camp closure.

Results

- 5.1** The results in this section were derived from the GRASP model running at a granularity fraction⁸ of $frac=100$. This is largely due to the computational constraints alluded to in Supplementary note 4. While some deviations will exist between GRASP predictions and empirical camp populations, this does not undermine the value of the model. Rather, it reflects broader limitations within the field of migration modelling, where perfect alignment with empirical data remains elusive due to the inherent unpredictability and data scarcity of forced migration scenarios.
- 5.2** Table 2 displays the parameters used in the model. They were derived intuitively but require validation with evidence from real forced migrants and Malian residents⁹. Performing any form of optimisation on this utility model through the use of model outputs would increase bias and lose focus of micro-processes. For $\beta_{x,i}$, similar to the findings in RefugeeSIM (Hébert et al. 2018), the parameter correlates with the age of the individual, with older agents perceiving higher friction on longer routes.

Parameter	Value
$\beta_{F,i}$	1
$\beta_{x,i}$	agent age / 3000
$\beta_{P,i}$	1/1000 (or 0 if not a camp)
$\beta_{\theta,i}$	10
$\beta_{D,i}$	1
$\beta_{N_1,i}$	1/100
$\beta_{C,i}$	1
β_0	150

Table 2: Parameters of simulation

Empirical comparison

- 5.3** Figure 5 presents the Tabareybarey camp in Niger and the Mbera camp in Mauritania, which differ in both their temporal prediction behaviour and population dynamics. Tabareybarey exhibits a pronounced over-prediction during the major displacement surge before gradually converging towards the empirical observations, suggesting that the camp may act as a temporary transit location within the simulated migration network. In contrast,

⁸Although large, it is a comparably small amount to the 1000 migrant agent Hébert (Hébert et al. 2018) employs.

⁹This could be conducted through surveys or qualitative interviews.

Mbera under-predicts the mid-simulation increase in refugees but converges closely to the final observed population. Although substantial relative differences remain (Table~[Table 1](#)), these should be interpreted alongside the model's ability to reproduce the timing of major migration events directly from conflict data. For humanitarian planning, identifying periods of rapid population increase may be as important as minimising relative differences alone.

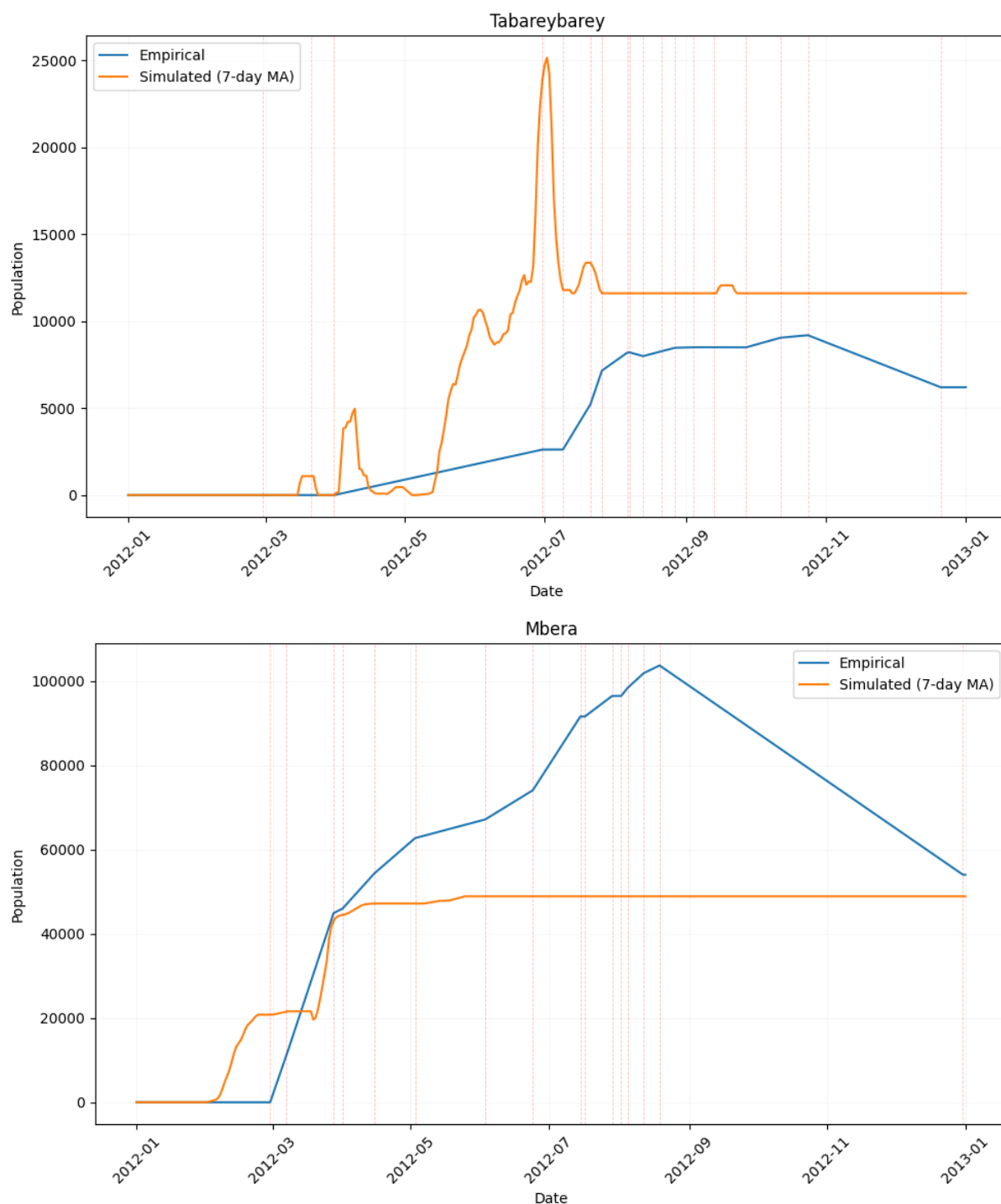


Figure 5: Empirical Comparison of GRASP results with UNHCR camp populations from Tabareybarey and Mbera

5.4 The $frac=100$ scale of GRASP results in low granularity, making the values sensitive to stochastic variation, particularly with the mean group size of 6 representing 600 agents. Larger camps tend to be predicted with greater accuracy than smaller ones, allowing for a reasonable estimate of the total number of refugees in camps (see Fig 6). The large spikes observed in the GRASP model likely stem from refugees using certain camps as interme-

diary stops on their way to final destinations. For instance, inner camps like Mentao, Goudoubo, Tabareybarey, and Mangaize serve as transitional hubs, connecting to outer camps such as Bobo, Ouagadougou, Niamey, and Mbera (see Fig. 2). This transition behaviour is particularly evident for Tabareybarey, where short-lived spikes suggest temporary accumulation of refugees before onward movement. Although smoothed versions of these outputs are available in the model outputs, the unsmoothed trajectories shown here more clearly illustrate the underlying model dynamics and the transition points between camps.

5.5 During model development, experiments were repeated at varying *frac* values to assess the effect of stochastic variation, with the results summarised in Supplementary Note 4. The final results presented in this paper were generated using a fixed random seed at the *frac* value described above to ensure reproducibility. While distributing the model across multiple processors or using parallel computing frameworks could, in theory, allow higher granularity runs, such methods were not implemented for the results presented here. The rationale and potential for high-performance computing approaches, including MPI-based parallelisation, are discussed in Section 6.12.

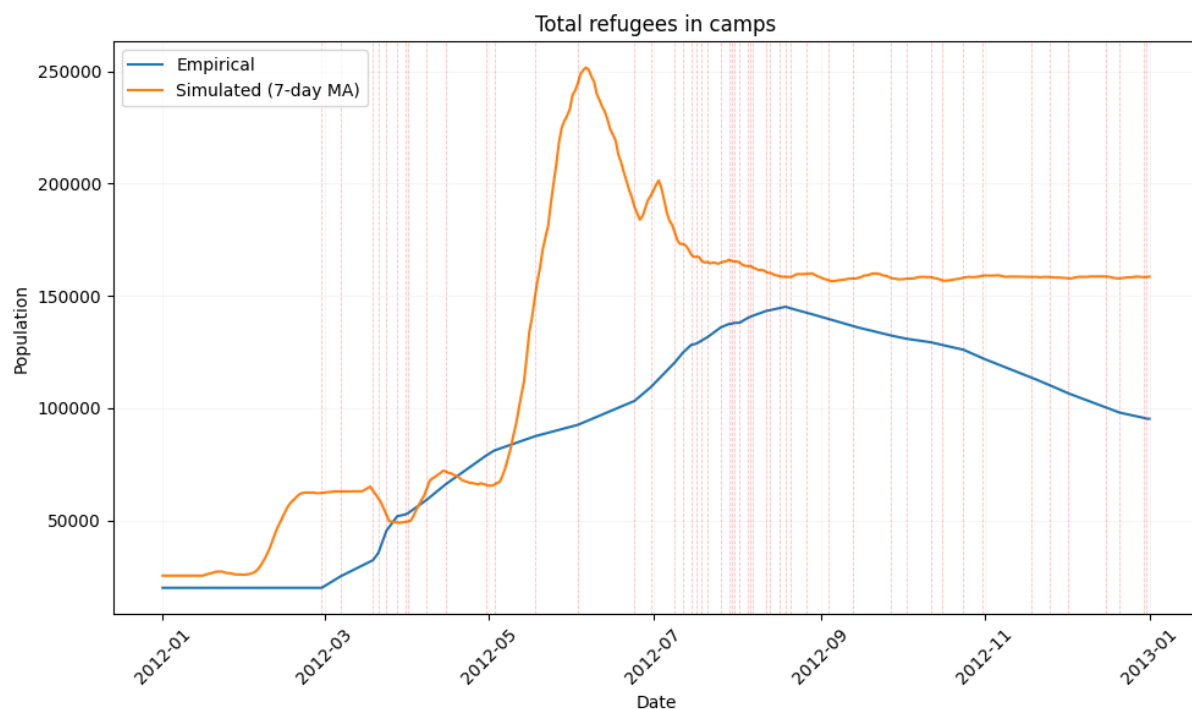


Figure 6: Number of refugees in camps

5.6 The main limitation of the GRASP model lies in its reliance on a calibrated utility function for destination selection. While parameter optimisation was performed at higher *frac* values due to computational constraints, this introduces uncertainty when the model is scaled to full-resolution simulations, as parameter sensitivity may vary with population size. Consequently, the utility function primarily affects the allocation of refugees between camps rather than the total number of refugees predicted. The utility-based approach nevertheless consistently outperformed the simpler alternative of selecting the nearest available destination, supporting its inclusion despite the remaining inaccuracies in camp allocation (Fig. 5). It is therefore important not to rely solely on aggregate model output for validation, as Cheong (2023) notes that this approach is insufficient. Instead, evaluating the underlying micro-processes is essential, motivating the incorporation of qualitative data from forced migrants to better inform the utility-based decision mechanism.

5.7 The displacement status (Fig 7), referring to whether individuals were refugees in camps, internally displaced persons (IDPs), or returnees, was largely predicted before April 30th, with a spike in IDPs attributed to numerous conflicts on that day. The UNHCR cannot obtain data on IDPs immediately after conflicts, and the limited span of asylum status data constrains further validation. Nonetheless, the proportion of IDPs is approximately 60%, aligning with global statistics (UNHCR 2022a). The precision of camped refugees generally exceeds 80%, decreasing during transition peaks and tailing off towards the end. This decline is primarily due to uncounted forced migrants immediately following conflicts and the limitations in GRASP's return logic, as refugees typically leave camps only after direct/indirect conflicts or camp closures. The accuracy and precision of classifying

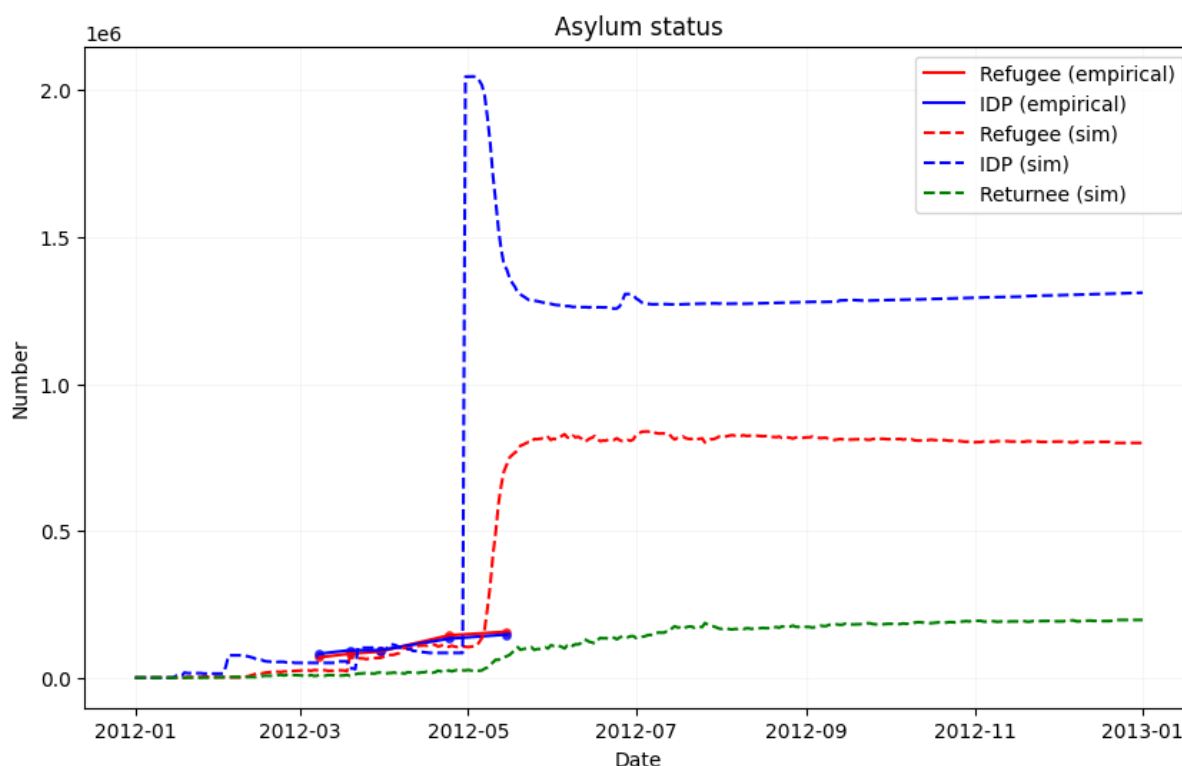


Figure 7: Displacement status

refugees in camps, total refugees, and IDPs are illustrated in Fig. 8, with the pooled classification metrics across the evaluation period summarised in Table 3.

- 5.8** These pooled metrics differ slightly from the average daily precision values reported elsewhere in the paper because classifications are aggregated across all empirical observation dates before calculating the performance measures.
- 5.9** Imprecision in IDP counts reflects the inherent challenges of displacement data collection, particularly the difficulty of capturing all IDPs and returnees, many of whom remain unregistered, are displaced multiple times, or move in ways that are not systematically tracked (Jacobsen 2006; Bradley 2020). These limitations affect both IDP and returnee figures, though they remain distinct categories.
- 5.10** Additionally, Fig. S6 in the Supplementary Information categorises the number of refugees for each camp, and all classification calculations were performed on the raw output of GRASP without prior scaling.

	Refugees (in camp)	Refugees (total)	IDPs
Accuracy	98.99%	98.84%	87.95%
Precision	79.56%	52.78%	79.23%

Table 3: Average¹⁰ accuracy and precision of agent status classification

- 5.11** Fig 9 predicts the flow of refugees by their country, indicating that Burkina Faso is the modal destination. However, there is limited empirical evidence from the UNHCR for comparison.
- 5.12** GRASP can also be used to identify individual paths, migrants settling in foreign settlements, and migrants travelling to airports. However, there is no current UNHCR data available to validate these aspects.

Flee comparison

- 5.13** GRASP has already demonstrated an improvement over the Flee model by predicting the total number of refugees in all camps with over 79% precision, a feat that Flee cannot achieve without relying on UNHCR registration statistics. The comparison below instead evaluates how accurately each model distributes those refugees between camps, which is primarily governed by the destination utility function. However, the distribution of

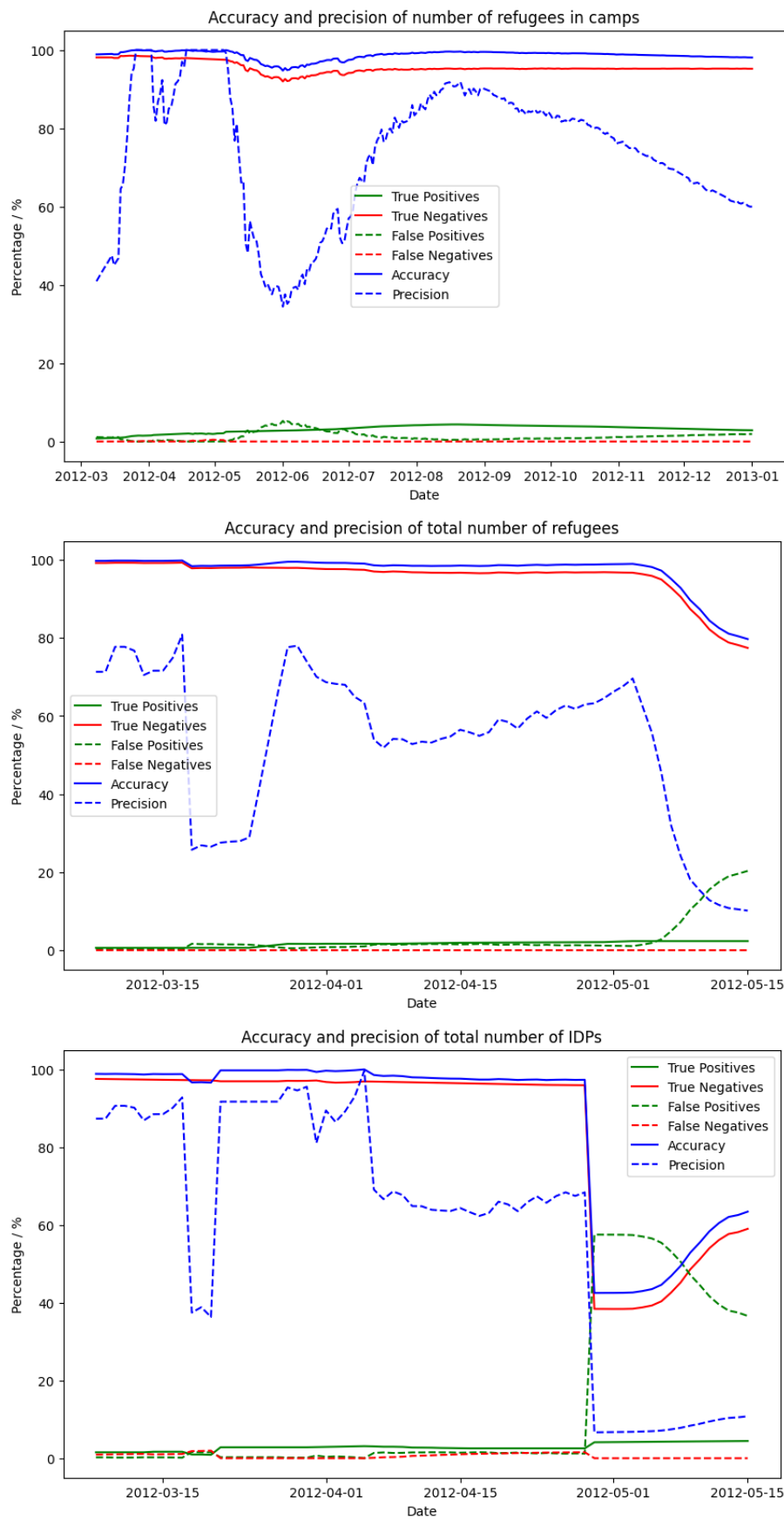


Figure 8: Accuracy and precision of classifying agent's status

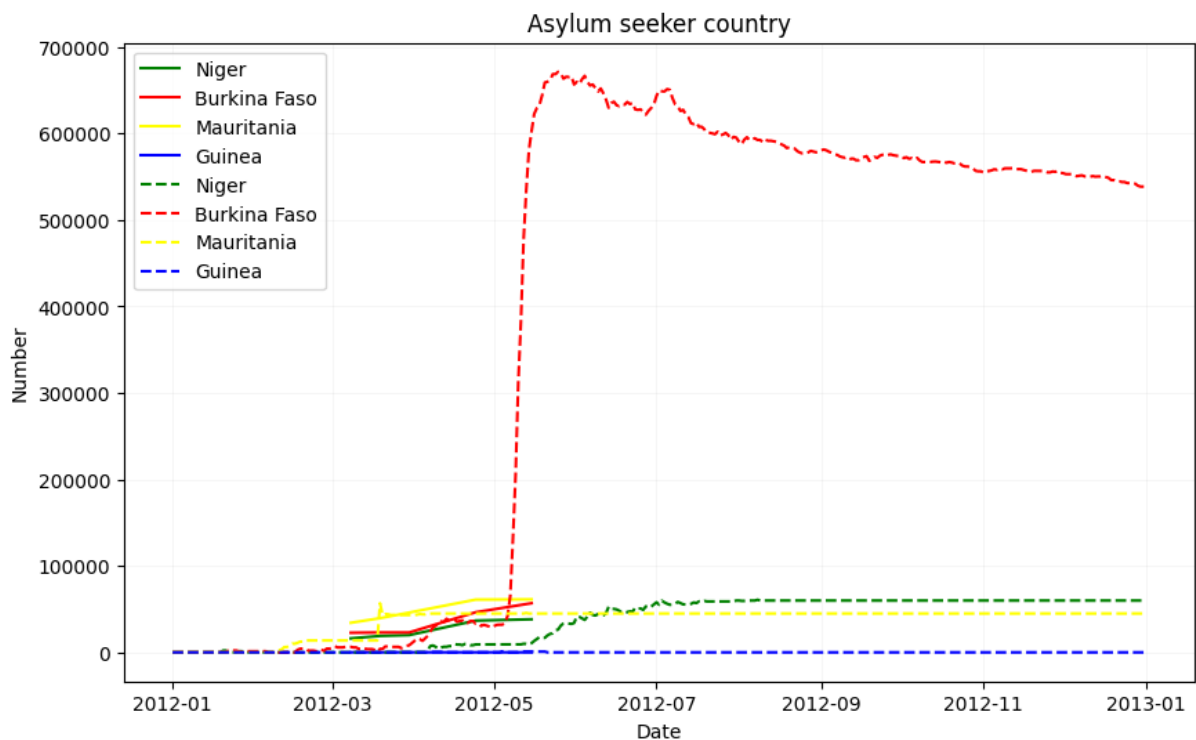


Figure 9: Asylum country

refugees among the camps can still be bench-marked against Flee. The Flee model was run using the intrinsic case study of Mali 2012. As previously mentioned, Flee scales its results by $N_{data,all}/N_{sim,all}$, where $N_{data,all}$ is the total number of refugees in camps from UNHCR on a particular day and $N_{sim,all}$ is the total number of refugee camps for the respective simulation (GRASP/Flee) on the same day.

5.14 Fig. 10 compares Flee and GRASP for Tabareybarey and Mbera (with all 7 comparable camps in Fig. S7 of the Supplementary Information). When scaled in the same fashion as Flee, it becomes apparent that GRASP is unable to predict forced migrant flow with a higher degree of precision than Flee when assessed solely by relative differences. However, GRASP more consistently captures the characteristic temporal behaviour of refugee movements, identifying the timing of major influxes, periods of population stability, and the onset of population decline while providing a reasonable estimate of camp populations. Large gradients are also detected, which are crucial for humanitarian support as they indicate when immediate aid is required for a camp. Regarding the characteristic shape of the forced migrant flows, Tabareybarey, Mbera, Abala, Mangaize, and Mentao all exhibit population trends that more closely resemble the empirical observations than those produced by Flee, even where discrepancies in population magnitude remain^{This is particularly important for humanitarian operations if refugee registrations are incomplete or delayed.}.

5.15 Figure 11 shows the difference in relative errors between the two models over time for Tabareybarey and Mbera, calculated with Equation 1. Fig. S8 of the Supplementary Information provides the evolution of relative difference for all camps.

5.16 For the majority of the simulation, Flee produced lower relative differences than GRASP. However, once the initial transition period had passed, GRASP frequently converged to comparable or lower relative differences for several camps, including Tabareybarey. Mbera exhibited a similar pattern, although Flee achieved lower errors towards the end of the simulation. Despite GRASP's ability to classify forced migrants and predict the total number of camped refugees, inaccuracies remain within the destination allocation logic, potentially arising from the transition peaks, utility function, or the implemented grouping and communication mechanisms. It is therefore essential to evaluate the contribution of these individual modules to both camp population distributions and forced migrant classification.

5.17 This distinction also highlights the importance of considering both short- and long-term predictive performance. In operational settings, models are continually recalibrated as new observations become available, making accurate short-term forecasts immediately following calibration potentially more valuable than long-term predictions alone.

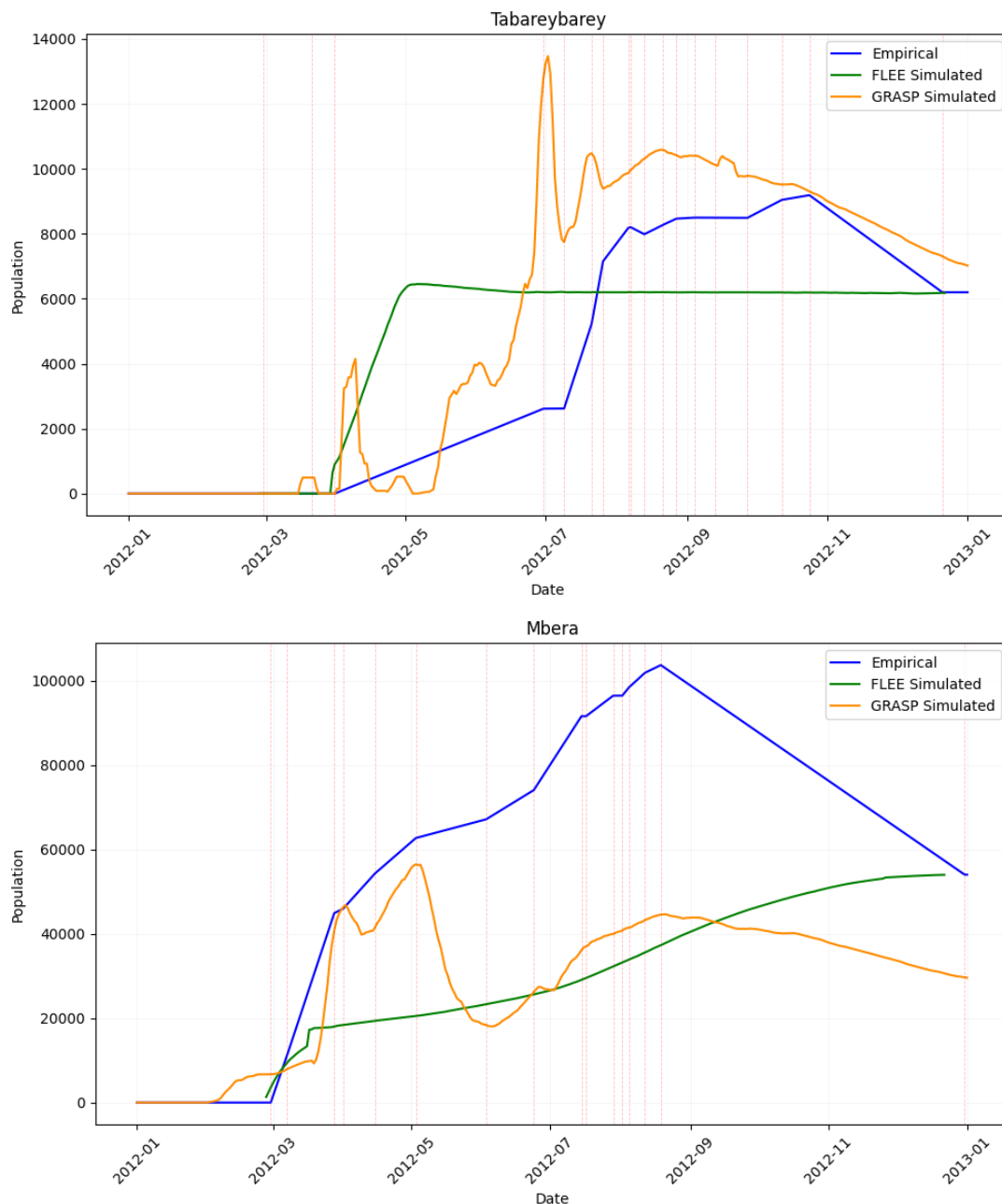


Figure 10: Comparison of GRASP, Flee, and UNHCR camp populations for Tabareybarey and Mbera

Feature Comparisons

- 5.18** Fig. 4 displays a table of the average relative errors (Eq. 1) of normalised camp populations for the inclusion and removal of features. The Flee average relative error is bench-marked. Fig. S9 of the Supplementary Information outlines the predictive behaviour of the scaled outputs of all models.
- 5.19** Although outperformed by Flee, GRASP achieves its lowest average relative difference when including the roulette function, grouping, and communication. Other than the roulette implementation, communication had the largest impact on the relative difference (approximately 55.7%) compared with grouping (approximately 33.7%). All results were batch-run three times and cross-averaged to avoid the impact of stochastic variation in the results. However, as indicated by Supplementary note 4, even at low *frac* values, the range of both precision and relative difference values varied greatly. There is an ongoing need for higher computation speed to increase the number of agents used and reduce model output variance.
- 5.20** While the average relative differences provide an overall measure of model performance, the individual camp trajectories reveal that each feature influences camps differently (Appendix). For example, communication

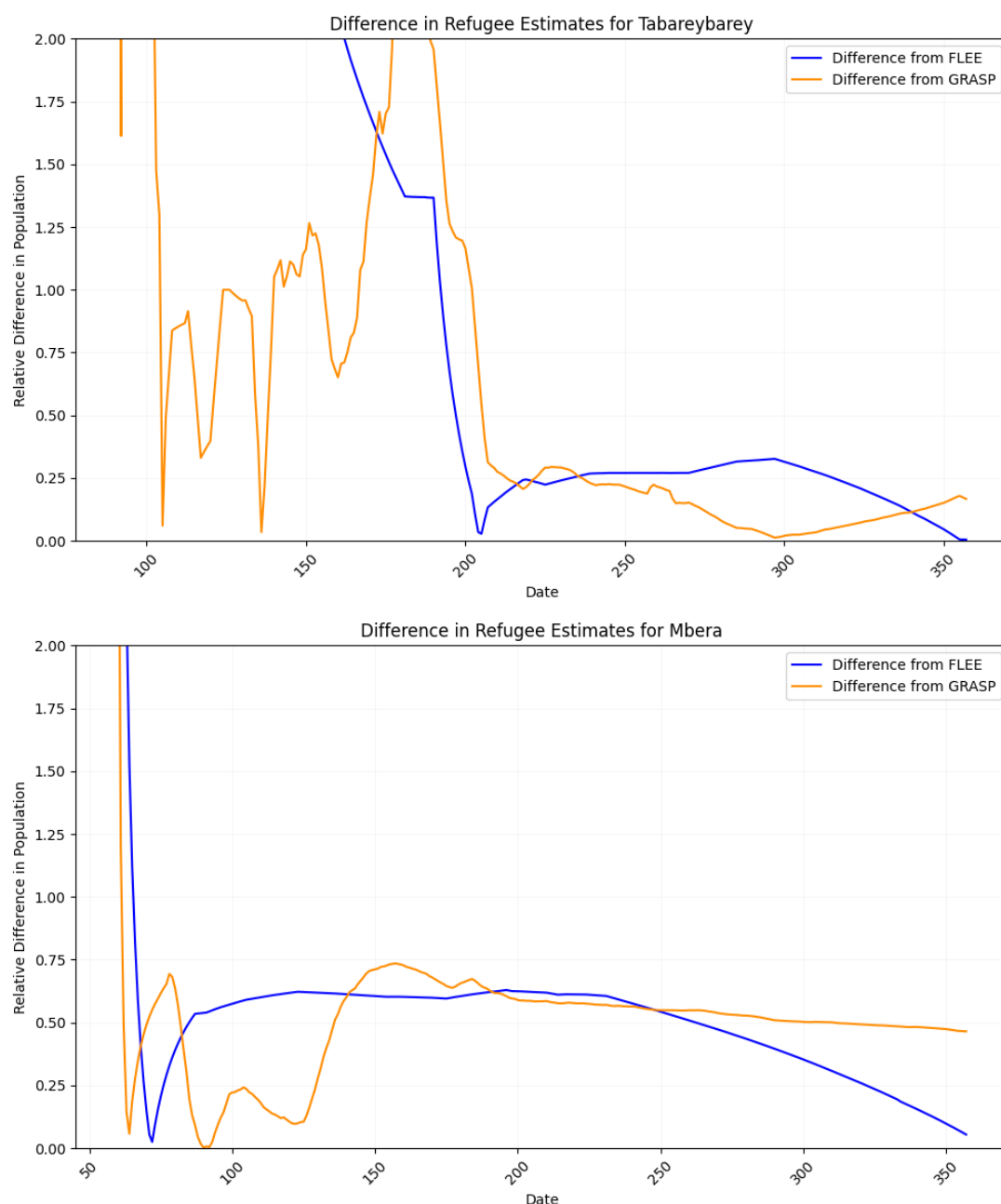


Figure 11: Relative difference of GRASP and Flee from UNHCR camp populations for Tabareybarey and Mbera

is essential for predicting refugee movements to Bobo, while removing grouping substantially increases the relative differences for Goudoubo, Mentaou, and Ouagadougou. Conversely, the roulette-based destination selection appears particularly important for Mbera, where removing it results in progressively larger errors over time. These differences suggest that model components influence both the short-term refugee surge and the longer-term redistribution of refugees between camps, indicating that feature importance is location-specific rather than uniform across the network.

- 5.1 In all versions of GRASP, the Burkina Faso camps of Bobo and Mentaou were largely overestimated, possibly indicating an unrecognised negative bias towards the destination country, perhaps related to economic or social ties that were not assessed within the scope of GRASP.
- 5.2 It is also important to assess the impact of features on GRASP's ability to predict the number of refugees in camps, the total number of refugees, and the number of IDPs. Table 5 outlines the average precision with which each of the models classify forced migrant status. Flee was not included due to the outputs being scaled up. The strongest argument for including grouping stems from the improvement in refugee classification precision. Removing grouping reduces the average daily precision of camped refugee classification from 77.85% to

	Flee	GRASP	GRASP without Grouping	GRASP without Communication
Bobo	48.91%	56.25	70.79%	177.52%
Mentao	19.85%	318.35%	358.50%	488.28%
Mbera	51.50%	37.19%	24.90%	41.54%
Abala	34.86%	87.35%	59.88%	82.50%
Mangaize	15.66%	191.21%	545.33%	322.58%
Niamey	80.24%	20.44%	32.37%	100%
Tabarey.	183.45%	114.81%	172.07%	42.05%
Total Diff.	434.47%	825.61%	1264.84%	1254.43%
Av. Diff.	62.07%	117.94%	180.69%	179.20%

Table 4: Relative differences of GRASP-predicted camp population distributions without selected features

29.97%, whereas removing communication reduces it more modestly to 69.62%. While communication provides a smaller improvement in classification precision, it substantially improves the spatial distribution of refugees across camps, justifying its inclusion despite the additional computational cost.

Model	Total precision
Flee	Unable to predict from ACLED data alone
GRASP	77.85%
GRASP without grouping	29.97%
GRASP without communication	69.62%

Table 5: Average daily precision of agents predicted to be in camps

● Discussion

- 6.1** GRASP is an ABM designed to predict immediate forced migrant flows after sudden conflicts, utilising empirical evidence on the grouping and communication of forced migrants. Although applied to Mali 2012 with UNHCR micro-data (Cheong 2023), GRASP can be extended to other cases using a combination of qualitative and quantitative data, including conflict event data (ACLED 2024), demographic structure (age and gender distributions), settlement and camp population data, and income inequality information derived from the World Inequality Dataset.
- 6.2** GRASP addresses limitations in current forced migration ABMs like the Flee 1.0 model, which fail to account for forced migrant experiences and micro-processes, leading to simulations with unverified fidelity. This research assesses the effectiveness of incorporating migrant topology (the interactive element of agents between themselves and their environment), focusing on grouping and communication, to improve forecasting in forced migration ABMs. These aspects were chosen based on advancements in literature (Hinsch & Bijak 2019; Bijak & Hinsch 2021; Collins & Frydenlund 2016). The dynamic weighting of nodes, based on evolving conflict data and border accessibility, also allows the model to reflect real-time constraints in migrant decision-making.
- 6.3** The release of Flee 3.0 represents a significant step forward in the field, with its more flexible API and ability to integrate social, demographic, and economic drivers such as food security and ethnicity (Ghorbani et al. 2024). However, GRASP offers complementary value. By design, GRASP abstracts from these heterogeneous attributes to focus on core behavioural micro-processes - namely, communication and grouping - allowing for a clearer interpretation of how such mechanisms shape displacement flows. In this respect, GRASP does not compete with Flee's expanded feature set but rather supplements it, providing a framework to interrogate fundamental behavioural rules underlying forced migration. This distinction ensures that even with improvements in Flee, GRASP continues to serve as a useful comparative model, offering explanatory power and interpretive clarity in addition to predictive performance.

- 6.4** The modular design of GRASP enhances its practical application by allowing users to adapt the model with minimal effort. This design principle is central to GRASP and represents one of its most significant contributions to the field. The code structure isolates key modules, such as agent movement, grouping, and communication, ensuring that these remain unchanged, while the main script serves as a customisable example for specific case studies. This flexibility makes GRASP highly accessible, as users can modify inputs such as regional data, fatality thresholds, or wealth distributions without altering the core mechanisms. Supplementary Note 3 highlights how the roulette-based decision-making method can be adjusted for specific contexts by refining utility functions to reflect regional preferences. Thanks to detailed comments within the publicly available code, users can quickly replicate the model for a new context, with basic configurations potentially taking less than a day. Furthermore, more advanced customisations, such as increasing the granularity of utility preferences or expanding the number of nodes, can be completed within 2–3 days, making the model both scalable and adaptable for diverse applications.
- 6.5** While parameters were sought empirically, some proxies were used and can be refined with further qualitative data similar to the interviews conducted by Cheong (2023). The utility function for scoring destinations can be parameterised with preferences from forced migrants or residents. Additional micro-data from UNHCR and economic data from the World Bank, World Inequality Database, and Oxfam were used as model parameters, with conflict data serving as input and the output validated against camp population data.
- 6.6** There is potential for trace validation of forced migrant paths and decisions produced by GRASP, performed by SMEs (Gore et al. 2017). Some other relevant outputs of GRASP that cannot be validated due to the gaps in data are the number of returnees, the number of refugees in foreign settlements (as opposed to camps), and the number of refugees who fled via an airport.
- 6.7** Minimising periods of under-prediction is critical for humanitarian organisations, such as the UNHCR, as insufficient aid can severely impact refugees' health, quality of life, and, in extreme cases, become life-threatening. Thus, while overpredicting might result in surplus resources, under-predicting poses a far greater risk to the well-being of vulnerable populations.
- 6.8** As shown in Fig. 12, GRASP generally predicts the total number of refugees in camps accurately across all days with all variants. It produces the highest degree of precision with all features included and remains largely over 70% average precision. It rarely overestimates (approximately 1 month duration), making it a fairly conservative estimator of camped refugee numbers.
- 6.9** It rarely overestimates (approximately 1 month duration), making it a fairly conservative estimator of camped refugee numbers. The temporary reduction in precision during the main displacement period reflects an over-prediction of the initial refugee surge, whereas the decline towards the end of the simulation is primarily due to an under-prediction of refugees leaving camps.

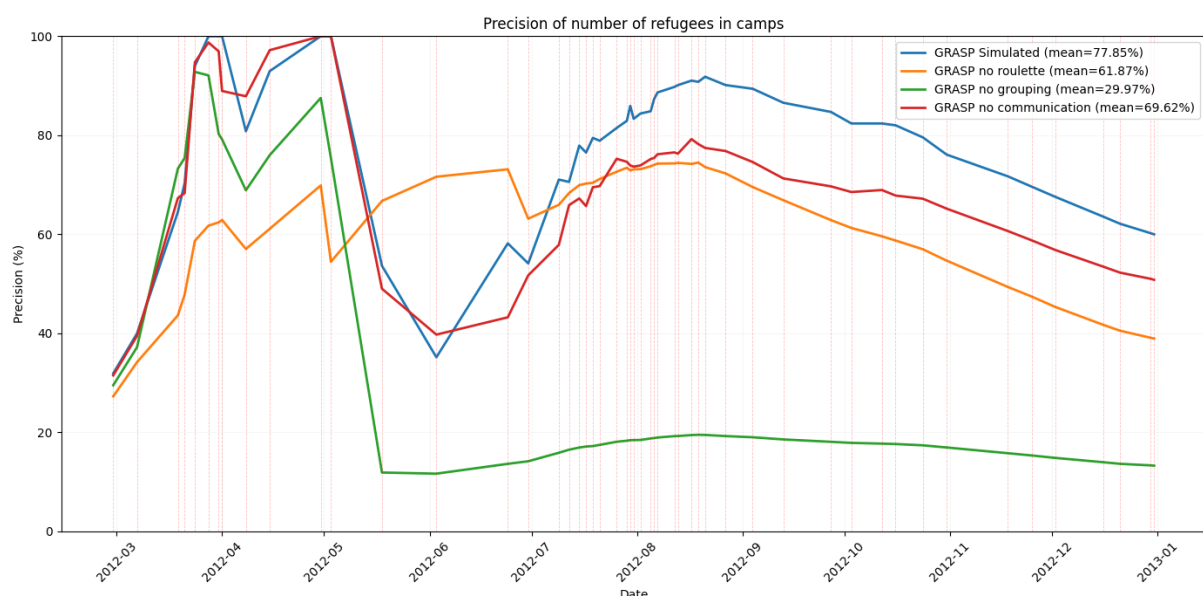


Figure 12: Precision of total number of refugees in camps

- 6.10** Accuracy and precision derived for the total number of refugees and IDPs are valuable for policy decision-makers as a tool for modelling responses to 'what if' scenarios such as border closures. The accuracy of the

total number of refugees out of all agents predicted was 98.99%, serving as a reliable tool for policy implementations. The IDPs, however, were only classified with 87.95%, which, as noted in Section 5 ??, reflects the inherent challenges of displacement data collection in conflict contexts rather than a critique of a particular dataset.

6.11 This research highlights the significance of incorporating grouping and communication in ABMs such as GRASP and Flee. However, the location selection model requires further calibration with qualitative data, as the utility function does not yet fully capture destination preferences. This primarily affects the distribution of refugees between camps rather than the prediction of the overall refugee population. This leads to the camp population holding a large relative difference of approximately 100%. Nevertheless, this difference was reduced when incorporating grouping and communication mechanics. The classification of refugees in camps was more precise with the implementation of all features. Future research should focus on integrating these elements into ABMs using empirically derived parameters for fatality thresholds, utility equations, and economic factors.

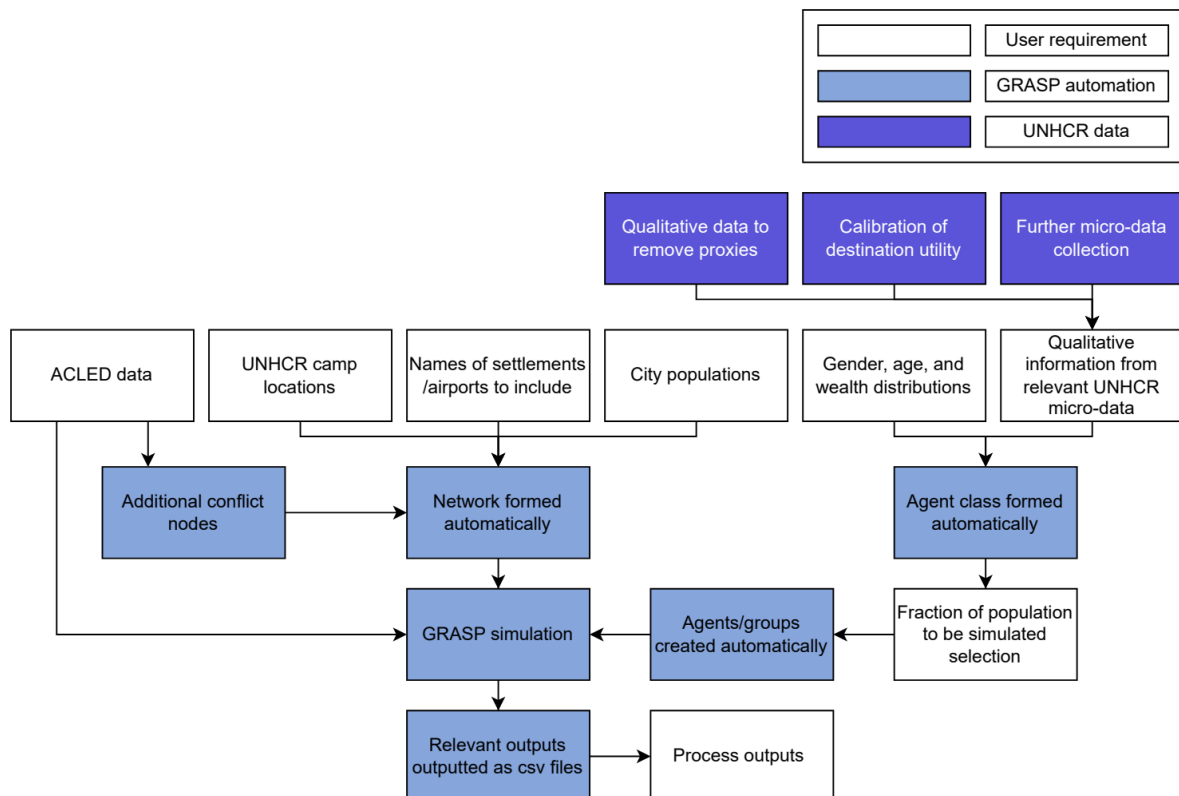


Figure 13: UNHCR contribution to GRASP

6.12 The results indicate that GRASP is a promising complementary tool for humanitarian planning rather than a direct replacement for established frameworks such as Flee. While Flee demonstrates strong performance in several benchmarks, GRASP offers additional capabilities through its representation of communication, social grouping, and heterogeneous decision logic, which provide a foundation for modelling the characteristic temporal behaviour of displacement beyond aggregate headcounts. The current implementation demonstrates the ability to estimate the total refugee population directly from conflict data while reproducing the characteristic temporal patterns of displacement. However, further development is required to improve the allocation of refugees between individual camps and fully exploit these behavioural features, particularly for capturing demographic composition and intra-camp dynamics where the existing population logic provides only preliminary groundwork. Future development should therefore focus primarily on improving the destination utility function, as this is the principal source of error in camp allocation rather than refugee count estimation. With additional validation and calibration, GRASP could support scenario exploration alongside existing models, helping agencies examine how information flows, social ties, and resource constraints shape movements over time. The following section addresses the limitations of the present study and outlines the work needed before such operational use.

Limitations

- 6.13** The major challenges in developing this model were related to data gaps in conflict zones and the need to balance model complexity with computational efficiency. In particular, the large number of agents meant that the final version of GRASP was computationally expensive, and simulations were run at 1% population, scaled by 100, with potential improvements through faster computing discussed.
- 6.14** It is important to note the intrinsic assumptions of GRASP, both when critically analysing its performance and identifying areas for future research.
- 6.15** In this paper, each agent represented 100 people. While most mechanisms are not statistically affected by this difference, the topological aspects of communication and grouping are inaccurate unless performed at full representation ($\text{frac}=1$). This is due to fewer agents interacting, leading to exaggerated group sizes and interactions. Additionally, this level of aggregation may introduce distortions in intra-family dynamics, as interactions that would occur within families are instead represented between agents. While optimisation of the code is crucial, with many agents, there is need for a much faster computer or cluster, or consultation for technical improvements such as Groen's implementation of MPI4py to parallelise Flee (Groen 2018), may be needed. However, any option with many agents is computationally expensive. A solution maybe to alter the topological aspects in a way that makes them more scalable and reflective of the full population under 1% of the simulation.
- 6.16** Secondly, proxies in the GRASP model need to be refined. For example, the agent's decision to leave is currently based on a fatality threshold derived stochastically by a uniform distribution (0,20). This distribution must be verified with empirical data. Another proxy is the speed of car travel and walking, which should be refined based on geography, road networks, and climatic conditions.
- 6.17** A key limitation of the current model is the absence of explicit economic push and pull factors between countries (both in the model performance and its comparison with Flee)—most notably, the pairwise income differences, which migration studies have consistently identified as strong predictors of cross-border flows. Wealth distributions are included at the agent level, though these are not related to macroeconomic disparities between countries. Future iterations of GRASP should prioritise integrating inter-country economic gradients into the destination utility function, especially when modelling transnational migration patterns. This requires empirical derivation or sensitivity analysis to determine the significance in an individual's decision logic.
- 6.18** It is also important to recognise that the broader field of migration modelling is still evolving, and theoretical predictions are inherently uncertain. Therefore, while GRASP may not consistently match empirical figures in all contexts, this reflects the complexity of modelling human displacement rather than a failure of the model. It also highlights the need to view predictive accuracy alongside explanatory power and policy relevance.
- 6.19** The grouping assumptions in GRASP may not reflect to all forced migration scenarios, e.g. evidence from the Lesvos, Greece 2015 refugee crisis shows that early migration waves may comprise predominantly single travellers, with different patterns emerging later (Crawley et al. 2016; United Nations High Commissioner for Refugees (UNHCR) 2015). This model does not currently simulate wave-specific demographic shifts, which may affect its applicability in such contexts. However, the model can partially represent dependencies in the selection of migration trajectories, insofar as the arrival of one agent at a destination can increase the likelihood that subsequent agents select the same destination.
- 6.20** Finally, despite the agent utilities proving more accurate than agents choosing the closest destination, the utility equation should be refined with data from forced migrants on their preferences for certain attributes. To address this, GDP can be implemented as a pull factor within the utility equation. GRASP's ABM nature allows for the implementation of any utility equation, storing agents' history and demographic data. As Frydenlund (2021) emphasised, simulation development should prompt data collection for higher granularity. Sensitivity analysis can be used for both model proxies and the parameters of the utility function to outline the impact of certain variables on refugee camps and boarder closures which will aid both operational decisions and 'what if' scenarios for policy makers.
- 6.21** A minor limitation of the network-based approach for ABMs is determining important nodes. Nodes with many links or intermediaries of many journeys tend to have higher populations than their capacity. Agents should exhibit a bias towards larger cities rather than smaller oversubscribed intermediary destinations. Also, GRASP only incorporates the Malian urban populations hence only considering urban forced migrants, though this maybe appropriate in model design given the lack of conflicts in rural locations.

Potential model improvements

- 6.22** While GRASP currently uses hard-coded geospatial entities for the Mali case study to allow for direct comparison with the Flee model, future implementations would benefit from linking the model to a dedicated geospatial database. This would support more flexible, scalable applications of GRASP to different regions, allowing the dynamic instantiation of nodes, links, and features such as border statuses or infrastructure from GIS layers or API sources.
- 6.23** In the current implementation, connections between nodes were manually defined to give the instance-builder full control over network granularity, especially when using a map. This approach makes link creation transparent and accessible, enabling tailored adjustment to specific case studies. While automated pairwise connection methods were considered initially, they quickly became infeasible as the number of nodes increased, resulting in overly dense and unrealistic networks.
- 6.24** The current model includes a limited form of memory, where agents retain and share knowledge of previously encountered conflict zones through a system of “no-go” nodes. This mechanism allows agents to avoid re-visiting dangerous areas and contributes to more informed route planning. The agent does not have global memory of its full trajectory, but instead operates with a local memory of its immediate past interactions and exposures. However, this memory is specific to conflict exposure and does not extend to a broader recollection of the agent’s full path or personal experiences during migration. A future iteration of this study may incorporate a longer-term memory that is incrementally populated as the agent migrates. Future enhancements could build on this foundation by introducing fuller path-dependency, where agents remember the sequence of nodes traversed or assign qualitative experiences (e.g., hardship, delay) to locations. This would allow agents to refine decision-making based on accumulated experiences, such as avoiding inefficient or distressing routes, even in the absence of direct conflict.
- 6.25** Another valuable area for development is improving the model’s usability and reproducibility for external researchers. Currently, GRASP is designed for research-focused customisation, but lacks full pipeline automation or an interface for input/output management. Future iterations could adopt a more modular structure where key inputs, such as geographic entities, population distributions, or conflict events, are loaded directly from CSV files or a linked geospatial database. This would enable researchers to instantiate new case studies by simply pointing to updated datasets, executing the model scripts, and generating outputs such as visualisations and statistics automatically. While not essential for this research-focused paper, making GRASP more plug-and-play would significantly enhance its accessibility and scalability for broader use across migration studies.
- 6.26** Additionally, it can be argued that conflict alone is an incomplete predictor of cross-border displacement, as many affected populations delay movement or prioritise staying or internal relocation depending on perceived risks, resources, and social ties (Hagen-Zanker et al. 2024). Although GRASP has the ability to build upon risk perception, with agents created from the offset as residents rather than directly from conflicts, there would be a benefit to incorporating delays, resources, and social ties into both the initial decision to flee and the agents’ decision logic.
- 6.27** A useful extension would be a sensitivity analysis in which elements of the agent decision logic are systematically perturbed to evaluate how alternative behavioural assumptions affect predictive accuracy and model stability. A further practical enhancement would be the development of a lightweight GUI that allows aid planners to seed their own assumptions and local data, run scenarios without technical expertise, and rapidly compare outcomes for operational decision-making.

Future research

- 6.28** There are opportunities to enhance GRASP’s granularity by adjusting movement speeds based on agents’ ages and incorporating transport costs as an agent dis-utility, such as that provided in Metafora (Cheong 2023). Integrating climate-based scenarios, as seen in the Flee model (Jahani et al. 2021; Kniveton et al. 2012; Campos et al. 2019), could also be beneficial. This is due to the climatic stress factors mentioned in Section 3.2, including global warming, deforestation, desertification, soil erosion, and inadequacy of potable water.
- 6.29** As eluded to in Section 3.2, Mali exhibits varying terrain with the Sahel desert in the north. There is an opportunity to attribute distances, risks, and speeds to the geography of the land. This can include inaccessible links as well as node frictions to consider in the utility function when the agent decides their path. Section 3.2 also presents the prevalent issue in Sahel migration routes of human trafficking. This could likely be explored with a similar implementation of node/link based helpers to Metafora (Cheong 2023).

- 6.30** Regarding communication, further features such as trust, perceived safety and risk, and effective friction could be included (Bijak & Hinsch 2021). In terms of grouping, additional research into group theory could enable group splitting at each node. Both communication and grouping could benefit from sensitivity analysis to fine-tune parameters. Future developments in computational intelligence methods, such as data mining and reinforcement learning, could reflect realistic agent behaviour, provided they are justified and not driven by trends. In particular, reinforcement learning or gradient-based approaches could be explored to calibrate behavioural parameters (e.g. β weights) from observed or synthetic trajectory data. The goal is to balance the improvement in granularity with the increase in computational complexity.
- 6.31** It is important to note that the comparisons in this paper are made against Flee 1.0, as GRASP was developed prior to the release of Flee 3.0 (Ghorbani et al. 2024). While Flee 3.0 introduces adaptable rules for agent movement and creation, as well as refined mechanisms for incorporating food security, ethnicity, religion, gender, and age dynamics, GRASP remains independent of these dimensions. In particular, age is treated differently across the two models and has been found to have limited impact on displacement dynamics in this context. Thus, the comparative tests with Flee 1.0 remain valid. At the same time, GRASP provides valuable insights by targeting a different scope: emphasising micro-processes of grouping and communication. That said, the improved API and modularity of Flee 3.0 highlight important progress in the field, and future research may explore benchmarking GRASP against later versions.

● Conclusion

- 7.1** Forced migrant movements, and the subsequent provision of humanitarian aid, are one of the greatest challenges facing societies across the world. Agent Based Models (ABM) can be used to forecast forced migrant movements. A very widely recognised ABM is the Flee model. While it has many advantages, it does have notable limitations that hinder its effectiveness in certain contexts. This paper presents an improved ABM of forced migration movement by incorporating grouping, communication, and higher granularity in agent decision logic. Based upon a critical literature review on forced migration forecasting, a methodology is developed for an ABM to complement the state-of-the-art Flee model. A detailed explanation of the Grouping and Rumours of Asylum Seeking Persons (GRASP) model architecture is further provided. To highlight its efficacy, this paper reviews GRASP's output at a fractional scale (one agent per 100 individuals) for the Mali 2012 case, comparing it against United Nations High Commissioner for Refugees (UNHCR) data and the Flee benchmark, while also discussing computational performance. The comparison between Flee and GRASP is analysed, highlighting GRASP's limitations and opportunities for future research.
- 7.2** GRASP predicts refugee destinations, including foreign settlements, UNHCR camps, and airports. GRASP also allows for trace validation of refugee movements by Subject Matter Experts (SMEs).
- 7.3** Key findings indicate that even at a 1% population scale, GRASP outperforms the Flee model in several ways. It is able to estimate the number of refugees in camps directly from conflict data alone (a capability that Flee lacks without relying on refugee registration data). It also identifies sudden gradients, peaks, and periods of stability in UNHCR camp populations, and classifies not only refugees but also IDPs and returnees with greater precision. Additionally, GRASP utilises empirical data for the micro-processes involved in refugee movement where possible, rather than relying on assumed proxies, and provides a utility-based decision model that can be further refined through interviews with Malian residents and refugees. Finally, it is capable of estimating the number of non-camping refugees, including those seeking refuge in neighbouring settlements and those fleeing via airports.
- 7.4** This paper demonstrates the crucial importance of communication mechanisms which have both a minor impact, on increasing model precision, and a major impact on the camp population distribution. Grouping mechanisms however significantly improve precision and camp population distribution. This feature has been included for higher accuracy. The large relative difference in predicting camp destinations individually, through camp population distributions, indicates that the utility model is currently inadequate for camp selection (and hence likely inadequate for destination selection). The utility model did however yield higher precision, whilst reflecting natural human behaviour as outlined in the Literature review and Methodology. Further research is needed to calibrate the utility function through interviews with forced migrants and crowd-sourcing information. The UNHCR can assist with this process. This will, in turn, improve the accuracy of GRASP and form a stronger argument for the implementation of grouping and rumours into future forced migrant ABMs.
- 7.5** The significance of this paper lies in that it both advances the understanding and modelling of forced migration. By identifying and implementing key mechanisms that enhance model precision, this research contributes to more effective humanitarian response strategies and policy-making.

● Appendix A: Overview, Design concepts, Details (ODD)

The model description follows the ODD (Overview, Design concepts, Details) protocol for describing individual- and agent-based models (Grimm et al. 2006; Grimm et al. 2020).

Purpose and Patterns

Purpose

The purpose of **GRASP** is to assess whether minimally-computationally-intensive grouping and communication mechanisms, embedded in a rapidly configurable ABM (GRASP), are sufficient to predict camp populations and their temporal dynamics arising from violence-driven forced migration.

The operational (higher-level) purpose is to provide a model that can be rapidly constructed and calibrated for new humanitarian crises to support situation awareness and short-horizon forecasting (e.g., camp sizing, supply pre-positioning, and surge planning).

GRASP aims to address the following question:

“Do rules for group movement and information sharing, coupled with conflict inputs, and travel constraints, reproduce observed camp-level arrivals and occupancies with operationally useful accuracy?”

The following are valid outputs that can be used to judge success:

- Time series of predicted vs. observed camp populations (overlay and error metrics).
- Distribution of arrivals across camps (cross-sectional fit per reporting period).

The model is considered to successfully employ group movement and information sharing for a case study if it achieves an improvement of 10% on both the above outputs, over the evaluation period of simulation (recommended a year minimum).

Patterns

Below are concise, testable patterns used to design and evaluate the model. Each pattern includes how it is tested and how results inform usefulness.

P1. Violence–displacement monotonicity **Statement:** For a given origin area and window, higher violence intensity (e.g., conflict events, fatalities, or severity index) is associated with higher displacement outflows within a short lag.

Test: Compute association between violence metrics and subsequent arrivals (e.g., lagged cross-correlation or rank correlation) at lag $L \in$ average distance [km] from centre of country to camps / average speed of agents (days); expect positive association above threshold 50%.

Use: Failure suggests errors in network perception/push mechanisms.

P2. Distance/impedance decay **Statement:** Conditional on leaving, arrival probability at a camp declines with effective travel distance.

Test: Fit arrival shares against generalised distance; expect negative elasticity.

Use: Failure indicates route network or mobility rules are inadequate.

P3. Capacity saturation **Statement:** Camp arrivals slow as capacity/utilisation approaches a threshold, producing spillovers to nearby camps.

Test: Detect diminishing marginal arrivals once occupancy $\geq 90\%$ and reallocation to alternates within 1 to 2 degrees of separation.

Use: Failure implies capacity effects are missing or thresholds mis-set.

P4. Batchiness sensitivity sweep **Statement:** Increasing batchiness (the fraction of the population simulated, `frac`) may raise error and dispersion, especially without grouping.

Test: Run ensembles at different batchiness levels (0.2–20%) with grouping on/off, measure MAPE, CRPS, and dispersion indices.

Use: If grouping keeps $\text{MAPE} \leq 25\%$ and $\text{CRPS} \leq 0.25$ while reducing dispersion, the model is considered valid and grouping necessary.

P5. Communication increases concentration **Statement:** Information sharing shifts flows toward a subset of camps with better perceived conditions/connectivity, improving distributional fit.

Test: Compare camp arrival shares and prediction error with communication ON vs. OFF; error reduction $\geq 5\%$.

Use: Failure implies information channels are too weak/strong or mis-scoped.

P6. Transit-hub behaviour **Statement:** Some camps act as temporary hubs with above-average through-flow (arrivals followed by onward movement), especially near transport junctions/borders.

Test: Identify nodes with high turnover ratio and onward flow within 3–5 days; expect small set of hubs.

Use: Absence suggests route network or stay-duration rules need revision.

Entities, State Variables, and Scales

Entities

The table of all entities considered in grasp are outlined in Table 6.

Name	Represents	Cardinality	Why included	Notes
Agents	Individuals deciding whether / where to move.	[1, frac]	Capture heterogeneous movement decisions, grouping, and information effects.	Frac is a variable assigned at the start in an effort to improve computational performance.
Locations	Registered / known camps, cities, towns, airports.	1	Primary destinations and units for validation (population/arrivals), intermediary nodes, and the initial spawn point for agents	Formed from census data/UNHCR camp locations.
Travel Links	Roads / tracks / border crossings and associated impedances.	1	Constrain feasible paths and costs.	Haversine distances are used to avoid nodal variance and increased computational time, network is not fully connected with links to be assigned by the user.
Environment / Observer	Global clock and exogenous inputs (e.g., conflict timeline, reporting cadence, border policy, frac).	n/a	Manage global state and inputs not tied to a single agent.	

Table 6: Model entities

Agent state variables

The main state variables associated with the agents in GRASP can be seen in Table 7.

Variable	Meaning / Units	Type	Dynamic?	Range	Source / Initialization
Gender	Sex/gender identity of the agent	String	No	["Male", "Female"]	Scenario setup, census data inputted by modeller
Age	Age of the agent (years)	Integer (years)	No	≥ 0	Scenario setup, census data inputted by modeller

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Table 7 continued from previous page

Variable	Meaning / Units	Type	Dyna- mic?	Range	Source / Initiali- sation
Status	Displacement status	String	Yes	["Refugee", "Re- turnee", "IDP", "Resi- dent"]	Scenario setup; updated by events such as movement and conflict.
Fatality thresh- old	Propensity to flee set- tlement under a threshold of deaths	Float	Yes	≥ 0	Randomly as- signed at sce- nario setup
Group	The strate- gic/family group an agent belongs to	List of Agent objects (or None)	Yes	Variable length or None	Scenario setup; updated by group dynamics
Is leader	Whether the agent is designated as group leader	Boolean	Yes	{True, False}	Scenario setup, selected randomly if age lies in independent years (18-65)
In strate- gic group	Whether the agent is part of a strategic subgroup	Boolean	Yes	{True, False}	Scenario setup, randomly as- signed with probabilities taken from UN-HCR microdata
Capital bracket	Economic capital category	String	No (or Yes if al- lowed to change)	{"Rich", "Medium", "Poor"}	Scenario setup, socioeconomic distribution
Speed	Daily travel capacity (km/day)	Float	Yes	>0	Scenario setup; can vary with age
Contacts	Known social contacts outside the immediate group	List of Agent objects	Yes	Variable length	Scenario setup as empty list; up- dated as network evolves
Moving	Whether the agent is currently travelling	Boolean	Yes	{True, False}	Updated each time step
Travel time	Duration of the cur- rent trip (number of days)	int	Yes	≥ 0	Updated during movement

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Table 7 continued from previous page

Variable	Meaning / Units	Type	Dyna-mic?	Range	Source / Initialisation
Is stuck	Whether the agent is unable to move (based on perception of network)	Boolean	Yes	{True, False}	Updated when becomes stuck and Agent is deemed no longer able to move.
Distance travelled since rest	Accumulated km since last stop	Float (km)	Yes	≥ 0	Initialised at 0, updated during movement
Location	Current position of the agent	Location object	Yes	Any valid Location	Updated as agent moves
Long-term destination	Intended final location	Location object	Yes	Any valid Location	Whilst agent is moving, changes at each step
Perception of the network	Agent's internal map of routes	Network X graph object	Yes	Graph structure	Initialised as subset of full network; updated with information gain and speeds

Table 7: Agent state variables

Location node state variables

It is worth noting the personalisation the user has in building the Network. However, it is important to ensure that above all else, all the locations in the ACLED database are included in your network. There were instances in the case study of Mali, where conflicts were needed to be mapped to the nearest available location node in the network.

Table 8: Location node state variables

Variable	Meaning / Units	Type	Dyna-mic?	Range	Source / Initialisation
Is camp	Whether the location is a camp or a city	Boolean	No	{True, False}	Model setup, based on dataset
Name	Name of the location	String	No	Any text string	External data source (e.g., UNHCR, census), inputted on model setup
Country	Country in which the location lies	String	No	Valid country names	External data, inputted on model setup

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Table 8 continued from previous page

Variable	Meaning / Units	Type	Dyna- mic?	Range	Source / Initiali- sation
Popula- tion	Population size of the location	Float (people)	Yes (can change with migration / conflict)	≥ 0	Census/population data; updated during simulation by migration and birth/death processes
Top 20 pop- ulous city	Indicates if it's in the top 20 by population	Boolean	No	{True, False}	Derived from population ranking at initialisation
Latitude	Geographic latitude	Float (degrees)	No	[-90, 90]	GIS data (assigned by Python package)
Longi- tude	Geographic longitude	Float (degrees)	No	[-180, 180]	GIS data (assigned by Python package)
Has airport	Whether the location has an airport	Boolean	No	{True, False}	Model setup, user required to acquire this knowledge
Conne- ctions / neigh- bours	Other locations directly connected	List of Location objects	No	Variable length list	User chooses where to lay connections in Network. Balancing, computational speed (minimal connections) and realistic routing.
Currently has a conflict	Active conflict in location	Boolean	Yes	{True, False}	Updated via ACLED dataset, if a conflict has happened in this region over the past 10 days, it will be True
Last conflict date	Date of most recent conflict	DateTime	Yes	Valid calendar dates	Initially None, stored when conflict occurs at Node
Is open	Whether location is accessible (not blocked / closed)	Boolean	Yes	{True, False}	Scenario rules, best used for camp openings/closure that are exogenous
Average number of days rested	Mean rest duration for agents at this location	Integer (days)	Yes	≥ 0	Computed during simulation from agent histories

Continued on next page

Table 8 continued from previous page

Variable	Meaning / Units	Type	Dyna- mic?	Range	Source / Initiali- sation
Fatalities	Number of deaths occurring at this location	Integer	Yes	≥ 0	Taken from ACLED database
Members	Agents currently present in the location	List of Agent objects	Yes	Variable length	Dynamic during simulation

Travel link state variables

Variable	Meaning / Units	Type	Dyna-mic?	Range	Source / Initialisation
Node 1	Name/ID of the first connected location	String	No	Any valid location identifier	From network definition
Node 2	Name/ID of the second connected location	String	No	Any valid location identifier	From network definition
Distance	Length of the connection between nodes (km)	Float (km)	No	>0	Derived from geospatial data (Haversine or GIS)
Bearing	Direction from Node 1 to Node 2	Float (degrees)	No	0–360	Derived from coordinates
Crosses border	Whether the connection goes across an international boundary	Boolean	No	{True, False}	Derived from country assignments of nodes
Number of agents travelled on link	Total count of agents that have used this link	Integer	Yes	≥ 0	Initialised at 0, updated during simulation

Table 9: Travel Link state variables

Environment / observer state variables

Variable	Meaning / Units	Type	Dyn-amic?	Range	Source / Initialisation
Global network	The overall network of all locations and links	Network X graph object	No	Graph structure	Constructed from user choosing what nodes to add and what nodes to link
Time step	Current simulation time	Date-Time	Yes	≥ 0	Starts at start point of simulation; increments a day each iteration
Borders open / closed	Global policy variable	Boolean	Yes	{True, False}	Scenario setup; based on information given on the conflict and user's discretion

Table 10: Table 5: Environment / Observer Variables

Process overview and scheduling

Schedule summary (per time step = one day)

At each time step, actions occur in fixed order: **Environment update**

- Simulation time is incremented by one step.
- New conflict events are updated from ACLED data.
- Global counters (population tallies, camp closures, return events) are updated.
- Environment state variables (settlements, camps, airports, road networks, conflict zones) are refreshed.

Conflict update

- Agents located in new or active conflict zones are displaced.
- Agents in camps affected by conflict are forced to move again.
- Return events are triggered if conflict subsides.

Group formation / adjustment

- Agents form or adjust family groups (fixed throughout journey).
- Strategic groups may form, merge, or remove the slowest members based on group utility.
- Group speed is updated according to the slowest member or group merging rules.

Communication

- Agents at nodes communicate with others (maximum of two contacts).
- Information shared includes destinations, “no-go” areas, and camp conditions.
- Contacts are stored; family members or prior acquaintances are notified when reaching camps.

Decision model (location choice)

- Each agent (or group leader) evaluates possible destinations using the utility function.
- Destination types include airports (rich), foreign settlements (middle), and refugee camps (poor).
- A roulette-based selection introduces stochasticity and bounded rationality.
- “No-go” nodes and received information filter potential options.

Movement

- Agents or groups advance along the network towards the chosen destination.
- Travel speed is determined by group composition (slowest member).
- If blocked (by conflict or closure), agents recalculate their destination in the next cycle.

Arrival check

- Agents reaching nodes are checked for camp entry, airport exit, or settlement arrival.
- Camp population counters are updated.
- Agent status is updated: refugee, IDP, returnee, or departed.

Figure ?? outlines the flowchart for agent decision making.

Rationale for Processing & Scheduling

- **Conflict before decision-making:** ensures agents respond to the most up-to-date security environment, avoiding outdated safe zones.
- **Group adjustment before movement:** guarantees group speed and composition are stable before travel.
- **Communication before decisions:** allows new information to influence destination choices realistically (rumours, contacts).
- **Decision before movement:** ensures each travel step is grounded in current options and updated group knowledge.
- **Arrival check last:** ensures camp populations and statuses are accurate at the end of each time step, providing correct inputs for the next cycle.

Design Concepts

Basic Principles At the system level, GRASP is designed to explore how violence-driven displacement varies with grouping and communication drivers and predicting camp populations with significant granularity. This model addresses a pressing problem in humanitarian response: anticipating the size and distribution of camp populations arising from violence-driven forced migration. Traditional approaches rely on statistical forecasting or detailed case-by-case analysis, but these are often too slow or inflexible for acute crises. GRASP differs by providing a rapid, generalisable agent-based framework in which camp population dynamics emerge mechanistically from simple but defensible rules of displacement, grouping, and communication. At the agent level, the model draws on the push–pull framework of migration and the concept of bounded rationality: individuals move primarily to reduce exposure to violence but do so using satisficing decisions constrained by imperfect information. The model therefore situates itself within both the humanitarian forecasting literature and agent-based studies of mobility under risk.

Emergence The key outcomes of the model are the time series of camp population sizes, the spatial distribution of displaced groups, and the timing and magnitude of surges into particular camps. These outcomes emerge from the interaction of agent-level processes, deciding whether to flee, choosing destinations, forming groups, and sharing information, combined with the spatial structure of violence and camp locations. Patterns such as the concentration of flows into certain hubs, lagged responses to violence spikes, and oversaturation of camp capacity are emergent rather than imposed, arising from decentralised decisions. At the same time, some structural constraints, such as maximum camp capacity or the fixed location of camps, are imposed features of the model environment.

Adaptation The agents' primary adaptive behaviour is the decision to flee violence and select a destination. This behaviour is modelled as direct objective seeking with satisficing: when violence intensity in an origin exceeds a tolerance threshold, households evaluate a set of potential destinations. Each destination is scored according to distance, perceived safety, and known services. Agents then select any option above an acceptability threshold rather than optimising globally, utilising a roulette method for probability of destination choices. The destination for each agent is updated at each timestep to allow the influence of drivers such as new conflicts, groups, communications, and border/camp closures. Strategic grouping is another adaptive process: agents may choose to join or remain within groups, which modifies both movement efficiency and the information they can access. Stochasticity in these choices reflects heterogeneity and imperfect rationality.

Objectives The implicit objective of agents is to minimise risk from violence while maximising the likelihood of reaching a safe and serviceable camp. This is expressed through a composite destination score (or utility) that balances proximity and safety against camp attractiveness, as mediated by communication. Unlike optimisation models, GRASP treats objectives as satisficing thresholds: agents are satisfied with any destination that provides "good enough" safety and access rather than always seeking the global optimum. This design choice reflects both empirical evidence of bounded rationality in displacement and the need for computational parsimony in rapid-deployment models.

Learning Learning is not implemented in the current model. Agents do not update their decision rules based on prior outcomes, nor do they accumulate experience across multiple moves. Instead, variation is represented through stochasticity and communication effects. This simplification is justified by the model's purpose: rapid short-term forecasting of camp populations, where displacement decisions are primarily reactive to acute violence rather than long-term adaptive strategies.

Prediction Agents do not make explicit predictions about future conditions such as worsening violence or camp closure. Instead, their behaviour reflects implicit short-horizon prediction: choosing to leave when violence exceeds a threshold assumes that movement will reduce risk, and choosing a destination assumes that perceived camp conditions will hold long enough for the journey. These implicit predictions are embedded in the satisficing rules but are not modelled as forward-looking optimisation.

Sensing Agents are assumed to know their own state (location, group membership, displacement status) and the violence intensity of their current origin. They also perceive information about potential camps, but this perception is imperfect and heterogeneously distributed: some agents may be aware of certain camps through communication links while others remain uninformed. Some may have out of date information. Camp distance is assumed to be accurately known, while camp attractiveness (e.g. capacity) may be distorted or incomplete. Group membership affects sensing: agents in larger or better-connected groups may have access to more information. This design reflects empirical evidence that displaced households often move with incomplete and network-mediated knowledge of destinations.

Interaction Interaction among agents occurs primarily through communication and grouping. Communication is represented as a form of direct interaction along social links: households share information about camp conditions, influencing each other's perceived attractiveness of destinations. Grouping is modelled as a mediated interaction: once agents join a group, their subsequent movement and sensing are affected by the group's properties (e.g., shared information, cohesion during travel). The range of interaction is therefore defined both by explicit social networks and by emergent group membership. These assumptions reflect empirical observations that displaced populations rely heavily on word-of-mouth and kinship ties to navigate uncertain environments.

Stochasticity Stochasticity is used in several places. First, agents' initial attributes (e.g., tolerance thresholds for violence, age, speeds) are assigned from probability distributions to introduce heterogeneity. Second, destination choice includes stochastic variation: when multiple camps have similar attractiveness scores, the final selection is randomised based on the proportion of their utility. Third, movement and grouping processes include probabilistic elements, such as whether a household decides to split from or join a group. These uses of stochasticity ensure that repeated runs generate different but plausible displacement trajectories, reflecting both unobserved heterogeneity in real populations and the uncertainty inherent in crisis situations.

Collectives Collectives are explicitly represented in the model as groups of displaced households and strategic groups. Groups influence member behaviour by constraining movement and information access. House group entities emerge dynamically as households choose to travel together but then persist as explicit entities that affect both agents and system outcomes. This representation reflects empirical evidence that many displaced people travel in family units or ad hoc collectives, and that such grouping shapes both the speed and direction of displacement. Strategic groups are assigned stochastically and are dynamically updated based on utility analyses.

Observation The key outputs observed from the model are camp population trajectories, the timing and magnitude of surges, and the spatial distribution of displaced households over time. These are collected as time series of camp occupancy and flow counts. Variability across runs is also observed, allowing uncertainty quantification. No “virtual scientist” submodels are implemented, but observation design is tied to the model's purpose: outputs correspond to the indicators most relevant for humanitarian planning, such as projected camp saturation within a specified time horizon.

Details

Initialisation Agents (individuals). A total of X agents are initialised (where X is the population/ $[\text{frac}]$ and $[\text{frac}]$ is to be chosen by the user), each representing an individual person. Each individual is assigned state variables (e.g., age, violence sensitivity threshold, gender, speed, capital class) by drawing from probability distributions reflecting empirical or literature-based ranges. Heterogeneity among agents is introduced through these distributions, so individuals differ in when they decide to flee, what routes they know, and how they respond to shocks.

Camps. A set of Y camps are created (where Y is chosen by the user based on available information/proposed new camps), each with initial attributes of geographic coordinates and baseline capacity. These values are either taken from case-specific datasets (e.g., UNHCR, IOM) or generic assumptions when applying the model to new contexts.

Origins. Individuals are initially placed at defined origin nodes representing settlements or districts. ACLED database is used in conjunction with individuals' fatality threshold to force the agent to flee.

Routes and networks. A spatial graph of possible movement paths between origins and camps is generated from a combination of GIS automation and manual tuning from the user. Travel distances are initialised via the Haversine equation (with travel times being heterogenous with age/capital bracket).

Groups. Family groups are assigned at the start. Dynamically these can grow and shrink with births and deaths respectively. Strategic collectives are also assigned based on empirical distributions. These groups are given a base tolerance towards the slow members of the group that gradually decrease (intolerance increases) when the group becomes stuck, eventually kicking the slowest member out on a utility based decision. Strategic groups can also merge/add new members. Each group has a leader to decide the destination and route.

Information layer. Each individual begins with an initial knowledge set of destinations. They learn of conflicts either directly (at their node) or indirectly (at neighbouring nodes) and communicate them with other members. They can communicate when they reach the camps to their family/contact lists. Each new piece of information affects the agents utility for each camp and affects their perception of the traversable network.

Spatial placement. Individuals are located at their origin nodes at $t = 0$. Camps and routes are defined with a network graph.

Case-specific vs. generic. Although Mali serves as a case study in this paper, initialization is not hard-coded to Mali. The architecture is designed to be rapidly deployable in new humanitarian contexts by plugging in available camp, conflict, and route data.

Scenario variation. Simulation scenarios may vary in initial conditions, such as:

- The location and capacity of camps.
- Population and census data (e.g. age, gender).
- Capital data (e.g. wealth distribution).
- The number and intensity of initial conflict events.

Rationale. Initialization is structured to capture heterogeneity at the level of individual decision-makers, as displacement trajectories depend on differences in perception and thresholds. This ensures that emergent camp population flows arise from individual-level variation rather than aggregate assumptions.

Input Data All data sources listed in Table 5 are publicly available and integral to the model design. The raw data builds on that used in the Flee model (Groen 2016), with additional data for improved granularity in the model implementations. The ACLED (2024) data offers details on conflicts in Mali from 06/01/1997 to the present, including date, type of disorder/event, actors involved, civilian targeting, region, geographic coordinates, and fatalities. UNHCR (2024b) reports provide information on camp locations and populations, border openings to Niger and Burkina Faso on 21st February 2012, and the closure of the Fassala camp on 19th March 2012. To compare with Flee, the model created uses data from 1st January 2012 to 1st January 2013, with temporal points linearly interpolated. Initial population values for the model on 1st January 2012 were extrapolated from UNHCR data. City populations for 2012 were linearly interpolated from the 2009 and 2022 censuses, assuming constant gender and age distributions from the 2022 census. Wealth distribution was estimated using data from The World Bank (2022), World Inequality Database (2022), and anecdotal evidence from Oxfam in West Africa (2023) about the upper 0.1% wealth.

Micro-data was extracted from UNHCR case studies on Sudan, Moldova, and Myanmar by (Cheong 2023). Household size distributions, though not directly related to Mali, are comparable to Sudan due to similar mean household sizes (The DHS Program 2020, ESRI 2023). Moldova and Myanmar were the only available sources for data on refugee travel groups and their sources of information.

Title	Source	From	To	Frequency	Relevant metrics	File format	Reference
Conflict data	Armed Conflict Location and Event Data (ACLED)	06/01/1997	Present	By event	Type, location, actors, fatalities	Published reports / website	(ACLED 2024)
Refugee data	United Nations High Commissioner for Refugees (UNHCR)	29/02/2012	31/01/2013	Varying	Destinations, camp populations	.csv	(UNHCR 2024b)
City populations	City Population	01/04/1998	2022/06/15	Three re-reported censuses	Populations for all cities in network across all countries	website	(City Population 2022)
Demographic Data	City Population	2022/06/15	n/a	n/a	Age and gender distributions	website	(City Population 2022)
Wealth distribution	World Bank	1994	2021	Six instances	Income percentile: 10%, 20%, 40%, 60%, 80%, 90%	website	(The World Bank 2022)
Wealth distribution	World Inequality Database	1995	2022	Yearly	Income and wealth percentiles: 50%, 90%, 99%	website	(World Inequality Database 2022)
Wealth distribution	Oxfam	16/01/2023	n/a	n/a	99.9% wealth percentile	website	(Oxfam in West Africa 2023)
Family size distribution (Sudan)	United Nations High Commissioner for Refugees (UNHCR)	2019	n/a	n/a	Household sizes	Published report	(Cheong 2023)
Group type distribution (Moldova)	United Nations High Commissioner for Refugees (UNHCR)	2022	n/a	n/a	Those travelling alone, with family, forming strategic groups, etc	Published report	(Cheong 2023)
Communication source distribution (Myanmar)	United Nations High Commissioner for Refugees (UNHCR)	2022	n/a	n/a	Those who trust family/friends with information	Published report	(Cheong 2023)

Table 11: Table 6: Data Sources and Attributes

● Appendix B: Supplementary notes

This section provides additional details on the model architecture and technical aspects of GRASP, along with further reasoning behind its design. It also serves as a guide for users to navigate the utilisation of GRASP and the analysis of its results.

To create or test models using GRASP, please access the repository from this link. The file *Case Study (Mali).py* should be used as the run file. Inline comments will guide users on which information to modify and where to input relevant data. For any questions regarding the code or its implementation, users can contact the developer by their email.

Additionally, we have included supplementary figures showing simulation results for all camps, beyond the two figures presented in the main text.

Supplementary note 1: GRASP movement mechanisms

The initial decision for an agent to flee is based on their fatality threshold, uniformly distributed between 0 and 20 (Stochastic variable used as a proxy. This needs verification with qualitative data from stakeholder interviews, such as refugees and residents who chose not to leave). Agents flee if the conflict in their region exceeds this threshold. Agents can rest during their journey by considering: 1) the number of days since the last conflict in the area, and 2) the distance travelled since their last rest. This prevents unrealistic simulations of continuous long-distance travel, while recognising the severity of the current zone. In cases of camp closures or immediate conflict, agents cannot rest.

Agent movement speeds are linked to their capital bracket, with poorer individuals walking at $u \sim N(200, 50^2)$ km/day and wealthier agents driving at $u \sim N(500, 100^2)$ km/day. Agents have their own perception of the network, filtering out links that are too long to travel in a day and nodes that are too dangerous to move through. Instead of committing to a long-term destination, agents continually reassess their destination based on conflicts and interactions. They move within their daily speed limit, often through one link, but faster agents in densely connected areas may travel through multiple nodes before resting. Their status is evaluated at each stage.

Similar to the Routes and Rumours Model (Hinsch & Bijak 2019), agents explore nodes both directly and indirectly while moving¹¹. This forms a set of *no-go nodes*, which are shared through communication between agents and used to filter out dangerous nodes when planning journeys. *No-go nodes* do not require a minimum fatality threshold and remain in the set regardless of when the conflict occurred. Camps with indirect conflicts (neighbouring nodes with conflict) are added to the set but can be removed 10 days after the conflict. If an agent is stuck with all neighbouring nodes inaccessible, they will remove all neighbours from the set, prioritising long-term success over short-term safety. This exploration logic is only applied to nodes where the agent resides, not intermediary nodes when travelling multiple nodes in a day. The *no-go node* logic is based on Hinsch & Bijak (2019), which states that agents migrate towards targets with limited initial knowledge, improved by exploration and social information exchange, impacting migration route dynamics.

Supplementary note 2: GRASP birth/death mechanisms

Unlike Metafora (Cheong 2023) and Flee (Groen 2016), agents are created pre-conflict as residents, so birth and death mechanisms are incorporated. Due to computational constraints, the simulation may have to run at a fraction (*frac*) of the population. The fatality rate provided by ACLED (2024) is divided by *frac* to create a probability of fatality for a member of the node in conflict. Depending on *frac*, the birth rate adjusts to either *N* births a day or *N* days per birth.

Supplementary note 3: Roulette selection method

In developing the GRASP model, which simulates forced migrant movements, the roulette-based decision model was implemented to enhance model precision. This method introduces significant uncertainty due to the lack of empirical calibration but is essential for improving predictive accuracy.

¹¹All neighbouring nodes within 50km are explored unless the agent is in a camp, in which case exploration is limited to 10km.

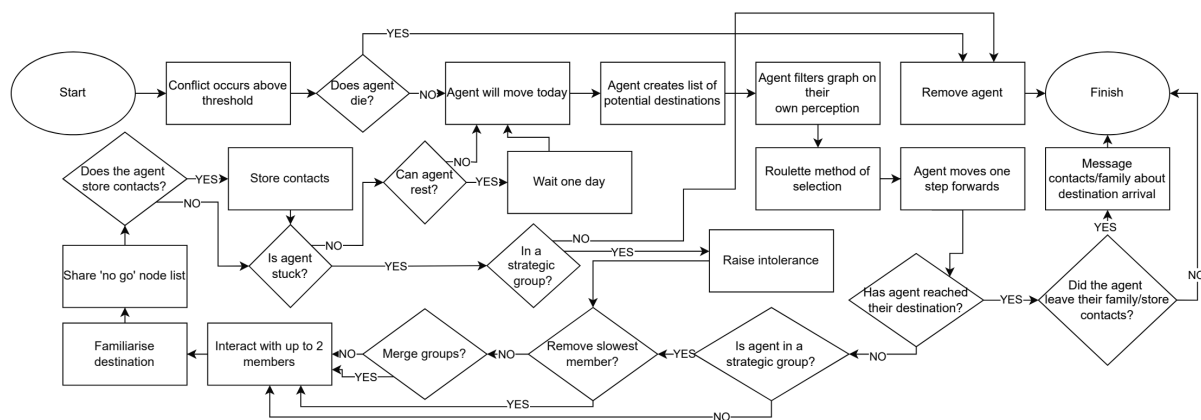


Figure 14: Simplified overview of agent logic.

Decision-making context Agent-based models (ABMs) often struggle to predict the irrationality involved in human decision-making (Bonabeau 2002). Forced migrants make decisions within the concept of 'bounded rationality,' where they weigh investments and risks for destination and route options (McAlpine 2021). Bounded rationality occurs under constraints, including cognitive limitations, where the decision-maker has a stable and well-organised order of preferences and the ability to compute which preferences are available to them (Dillon 1998).

Decision-making models Several decision-making models exist, including compensatory models that allow high performance on one attribute to compensate for low performance on another, and non-compensatory models that use screening or elimination of alternatives based on specific attributes. Holistic models consider each alternative in a non-sequential manner, whereas non-holistic models assess alternatives attribute-by-attribute (Simon 1955; Cohen et al. 1972; Beach & Mitchell 1987; Einhorn 1970; Tversky 1972; Klein 1989).

Planned behavior theory (Klabunde & Willekens 2016; Ajzen 1991), and other models considering risk, circumstances, and cost-benefit analyses, have also been applied in ABMs (Sandelowski 2000; Smith et al. 2010; Hébert et al. 2018).

Roulette selection method The roulette selection method allows the agent to consider the utility of each alternative over the sum of all utilities to derive the probability of choosing an alternative. This means the agent evaluates all alternatives proportionally to their benefits, including irrational choices, by adding a bias to each alternative (McAlpine 2021; Dillon 1998). This method significantly improves model accuracy, particularly in predicting camp populations.

Impact of removing roulette selection Removing the roulette selection function from GRASP simplifies the assessment of independent impacts of communication and grouping attributes but substantially reduces model accuracy. For instance, without the roulette function, agents choose the closest available option, which led to poor predictions for certain destinations like Mentao and Goudoubou due to their greater distances (see Appendix). This removal increased the average relative difference by 950% and reduced precision by 27%, highlighting the method's importance for accurate simulations.

Need for calibration Although the roulette method improves model precision, it requires empirical calibration with qualitative data to better represent the ranking of preferences and reduce inaccuracies in camp distributions (Bijak 2022; Bijak et al. 2019; Bijak & Wiśniowski 2011). Specifically, GRASP showed an improved relative difference in predicting camp population distributions with this mechanism, emphasising the need for further calibration to enhance its reliability.

In conclusion, while the roulette selection method introduces uncertainty due to the lack of empirical calibration, it is crucial for improving the precision and accuracy of the GRASP model in predicting forced migrant movements.

Supplementary note 4: Computational speed

The main limitation of GRASP lies in its complexity. Most simulations were performed on an Intel Core™ i9-9900K CPU @ 3.60GHz with 32.0GB of RAM. The *frac* value was adjusted within the simulation to compare run times.

GRASP is estimated to process 1% of Mali's population in approximately 40 hours, 10% in about 164 days, and 100% in roughly 45 years on this computer.

Impact of model resolution on precision and relative difference

There is an improvement of precision in predicting the number of camped refugees from the increase in computational run time (through increasing the percentage of the population being modelled - i.e., number of agents). Not only does the model get more precise with a lower *frac* value, it also exhibits a lower precision variance.

The increase in run-time also reduces the relative error associated with camp population distributions. This correlation however is less conclusive than the increase in precision.

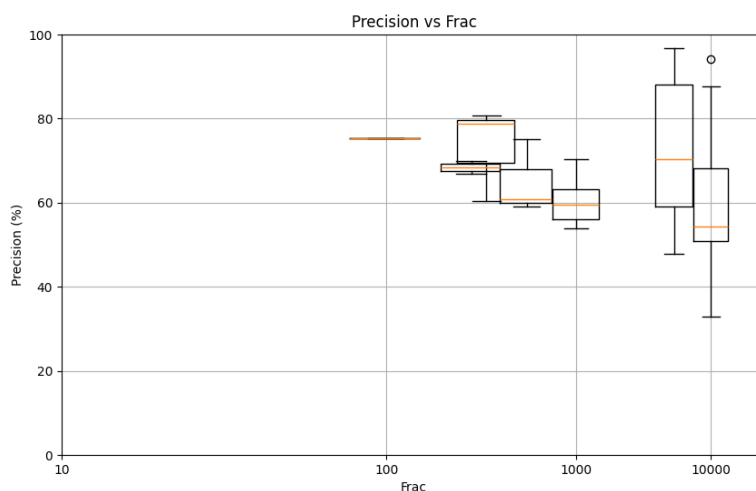


Figure 15: Camped refugee precision against fraction of the population simulated.

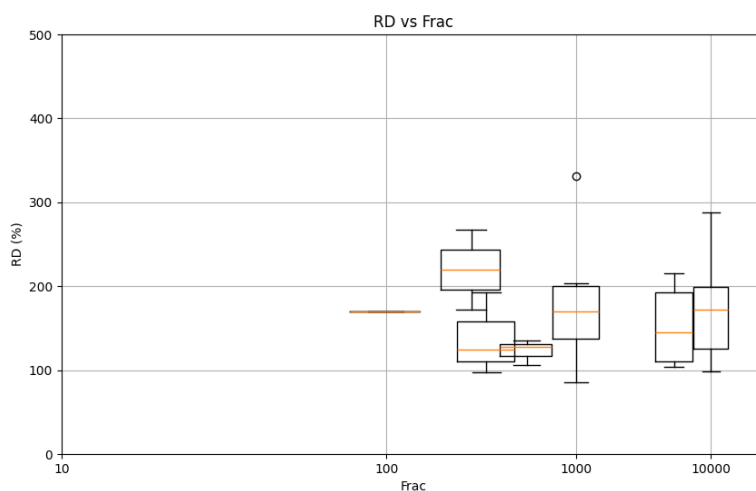


Figure 16: Camp population distribution relative difference against fraction of the population simulated.

GRASP could potentially benefit from running on a supercomputer or High-Performance Cluster (HPC), which would enable faster runs and more parameter iterations. Efforts were made during this project to utilise an HPC. However, due to the current high demand (June 2024) on Imperial's HPC and subsequent lengthy queues, this was infeasible during the short time-span required to obtain model results.

Impact of model resolution on computational speed

Two computers were utilised in both the model design and production of results. The *frac* values, f , were varied to assess the impact on computational runtime, T . The Apple™ Macbook Pro M1 was used for the majority of the model design.

With logged axes, the relationship is approximately linear with a gradient¹² of approximately -3.2. This indicates $T_{mac} \propto f^{-3.2}$. Given that $n = N/f$, where n is the number of agents in the model and N is the total population, with knowing $n \propto 1/f$, we can deduce $T_{mac} \propto n^{3.2}$. A remote computer was used for the results with Intel Core™ i9-9900K CPU @ 3.60GHz with 32.0GB of RAM.

The gradient on this graph is approximately -1.5. Using the same deduction we arrive at $T_{remote} \propto n^{1.5}$. With a lower n value (lower number of agents) the Mac performed at a faster speed than the remote computer. However, as n increased, the remote computer would increase in time at a lower rate. It was for this reason, that the Mac was chosen for model design and the remote computer was used for large simulations (obtaining results). It is also worth noting the remote computer has a larger memory than the Mac and so can handle excessive number of agents and attributes.

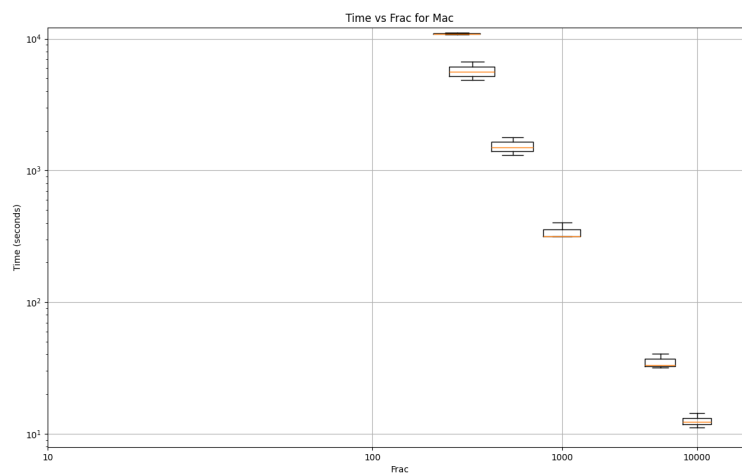


Figure 17: Apple™ M1 16GB run-time vs fraction of population simulated.

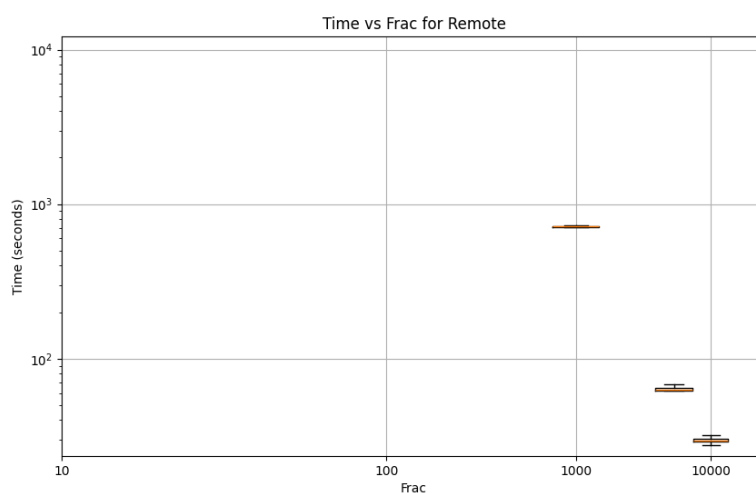


Figure 18: Microsoft™ Intel Core™ i9-9900K 32.0GB run-time vs fraction of population simulated.

¹²This value was arrived at from *scipy.optimize* fitting to the function $y = kx^a$.

Supplementary note 5: Understanding binary classification in the model

This Supplementary note is designed to aid users who are not familiar with confusion matrices in interpreting the binary classification results produced by the model. The goal is to provide a clear and user-friendly explanation of how the simulated population data are classified relative to empirical data.

In the context of this model:

- **True Positive (TP):** The count of individuals correctly predicted by the model to be in a particular state (e.g., within a camp) matches the empirical data.
- **False Positive (FP):** Occurs when the model predicts a higher number of individuals in a particular state than is observed in the empirical data.
- **False Negative (FN):** Occurs when the model predicts fewer individuals in a particular state than is observed in the empirical data.
- **True Negative (TN):** The difference between the total modelled population (Urban population of Mali + initial camp populations = 3,321,680) and the sum of True Positive, False Positive, and False Negative results. This represents individuals correctly identified as not being in the state.

These classifications allow us to compute critical model performance metrics:

- **Accuracy:** This metric reflects the proportion of total correct predictions (both TP and TN) made by the model out of all predictions made.
- **Precision:** This metric reflects the proportion of positive identifications that were actually correct (TP over TP plus FP).

Understanding these metrics is crucial for assessing the model's performance in predicting camp populations accurately.

This guide should help new users effectively utilise and interpret the model's output, ensuring that they can accurately evaluate the reliability of the simulation results.

Supplementary note 6: Further notes on validation methods

In the development of the GRASP model, which simulates the movement of forced migrants, our goal is to derive all proxies empirically. However, empirical data for certain parameters, such as the fatality threshold of individuals or movement speeds, may not always be available. This Supplementary note provides guidelines on compensating for these empirical gaps to enhance model validity and reliability:

1. Sensitivity Analysis (SA):

- **Local sensitivity analysis:** This method is employed to assess how changes in specific parameters, such as fatality thresholds, impact the model's outputs within a restricted input space. It helps identify the influence and sensitivity of individual parameters under controlled variations (Bijak 2022).
- **Global sensitivity analysis:** This analysis evaluates the impact of variations across the entire range of input parameters, providing a comprehensive understanding of the model's behavior under different scenarios (Bijak 2022). This is crucial for determining which parameters most significantly affect the model outcomes and therefore require careful approximation or further research.
- **Variance-based sensitivity analysis:** By examining how variances in inputs affect the variances in outputs, this method highlights critical parameters that significantly influence the model's behavior, guiding focused data collection and parameter estimation (Bijak 2022).
- **Sobol's indices:** These indices quantify the contribution of each input parameter to the total variance in the model's output. They are particularly useful for identifying which proxies have the greatest impact on the predictive accuracy of the model (Bijak 2022).

2. Compensatory data strategies:

- **Proxy data utilisation:** In cases where direct empirical data are unavailable, relevant proxy data should be considered. For instance, if direct data on movement speeds are lacking, traffic flow data or historical movement patterns during similar events could be utilised (Gilbert et al. 2018).
 - **Expert elicitation:** Engaging with subject matter experts (SMEs) can provide qualitative assessments that help estimate parameters based on expert knowledge and experience. This method is particularly useful for setting plausible ranges for parameters like fatality thresholds (Bijak & Wiśniowski 2011).
3. **Model calibration:** Calibration involves adjusting the model parameters iteratively to align simulation outputs with real-world data as closely as possible. This process compensates for the lack of direct empirical data by ensuring that the model's predictions are consistent with observed phenomena (Bijak & Wiśniowski 2011).
 4. **Uncertainty quantification:** Recognising and quantifying the uncertainty inherent in using non-empirical proxies is essential. Techniques such as Bayesian Uncertainty Quantification can be applied to provide a probabilistic assessment of the impacts of input uncertainties on the model's outputs (Bijak 2022).
 5. **Scenario analysis:** Conducting scenario analyses can test the robustness of the model under various hypothetical conditions. This helps to understand how changes in the non-empirically derived proxies influence the outcomes, providing insights into potential biases or limitations introduced by these proxies (Bijak & Wiśniowski 2011).

By employing these methods, users of the GRASP model can effectively address and compensate for the lack of direct empirical data, ensuring that the model remains a valuable tool for predicting the movement of forced migrants with a high degree of reliability and validity.

Supplementary note 7: Equation for Haversine distances

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (13)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (14)$$

$$d = R \cdot c \quad (15)$$

Where ϕ_1 and ϕ_2 are the latitudes of the two points in radians, $\Delta\phi$ is the difference in latitudes, $\Delta\lambda$ is the difference in longitudes, R is the Earth's radius (approximately 6371km), and d is the distance between the two points.

Appendix C: GRASP results across all camps

The main text saw the presentation of the results of GRASP on Tabareybarey and Mbera.

Fig. 19 displays the raw camp populations of GRASP. These values are not scaled or processed.

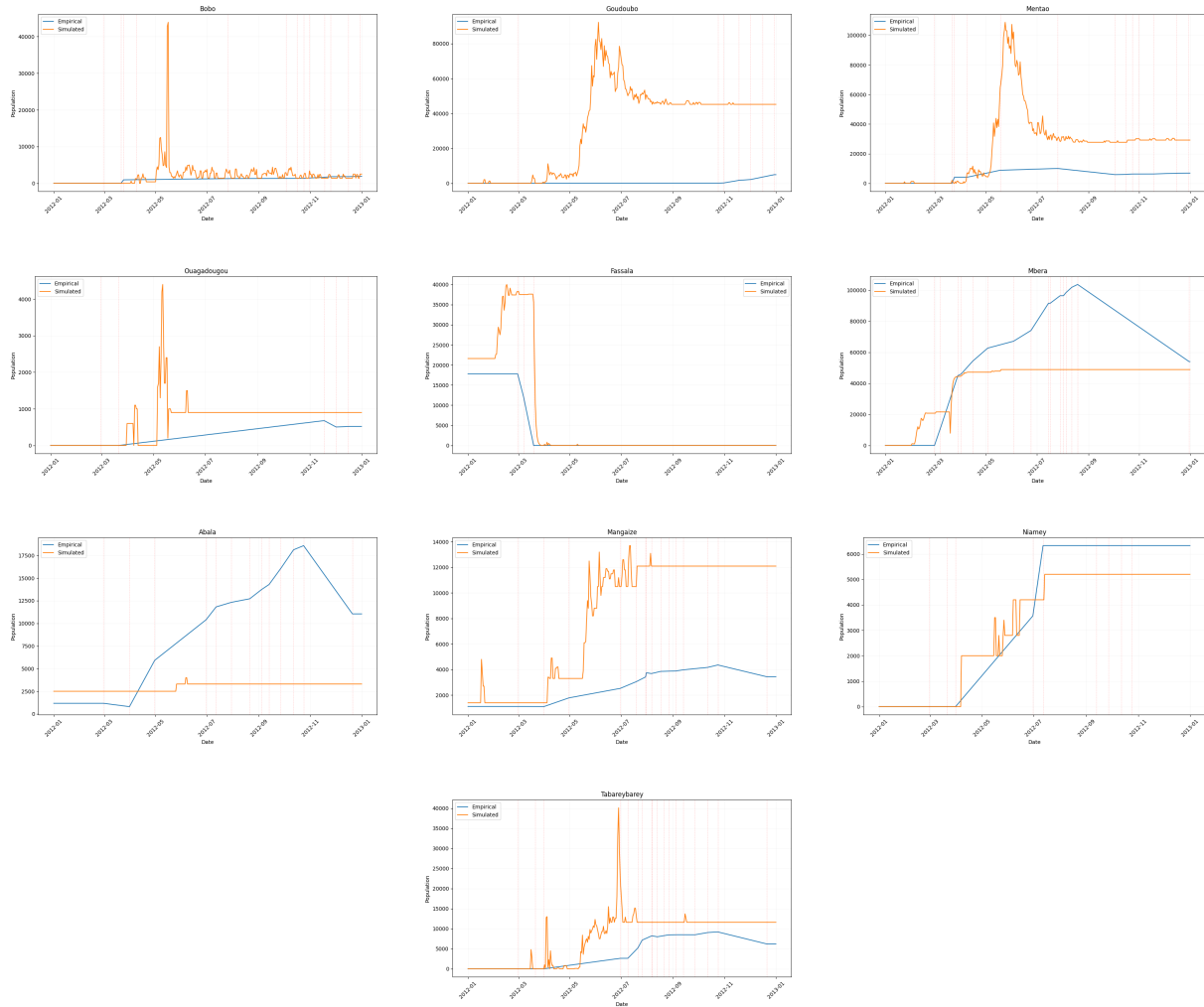


Figure 19: Empirical Comparison of GRASP results with UNHCR camp populations for all camps.

Appendix D: GRASP vs Flee for all camps

Fig. 20 compares the scaled output for camp populations for both GRASP and Flee.

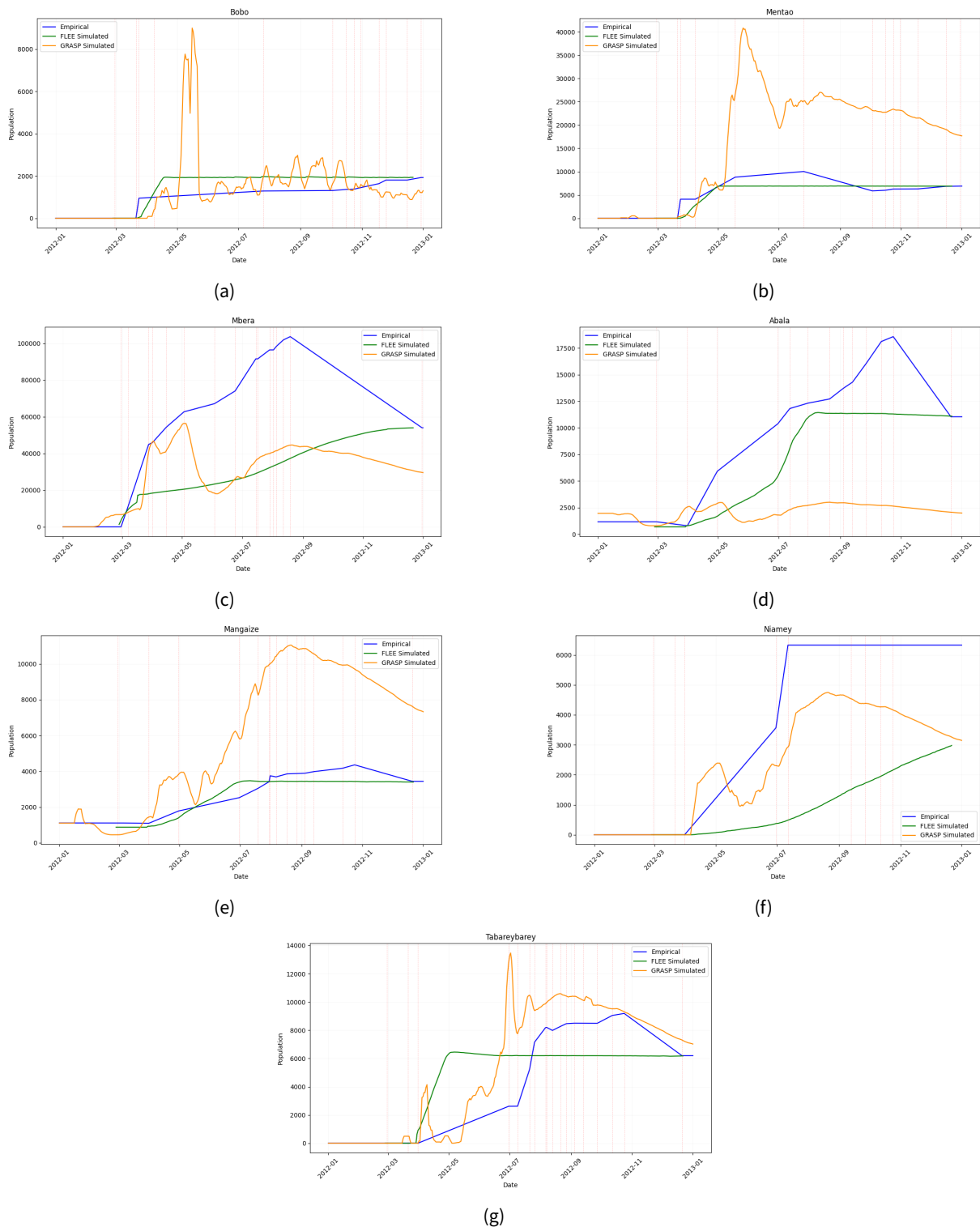


Figure 20: Comparison of GRASP, Flee, and UNHCR camp populations for all camps.

Appendix E: Relative difference (GRASP vs Flee) for all camps

Fig. 21 displays the relative difference between the scaled values of Flee and GRASP from the UNHCR camp populations as a function of the date.

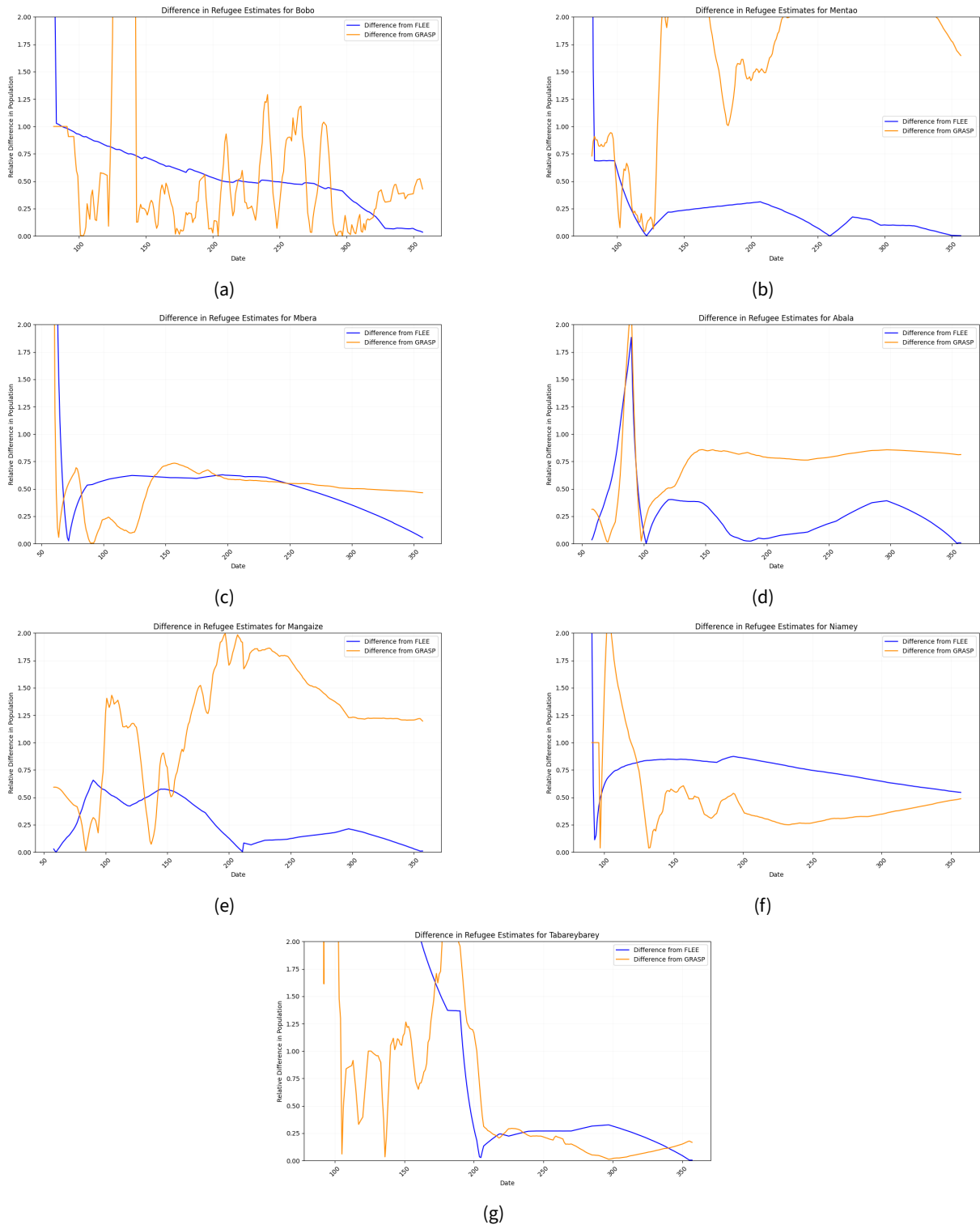


Figure 21: Relative difference of GRASP and Flee from UNHCR camp populations for all camps.

Appendix F: Forced migrant status classification for all camps

Fig. 22 shows GRASP's classification accuracy for agents in specific camps, based on raw, un-scaled outputs. The precision indicates how accurately GRASP predicts camp choices based on population distribution. The negatives are defined as refugees that are in camps but not in the particular camp in question. This is an entirely different statistic to predicting the total number of camped refugees.

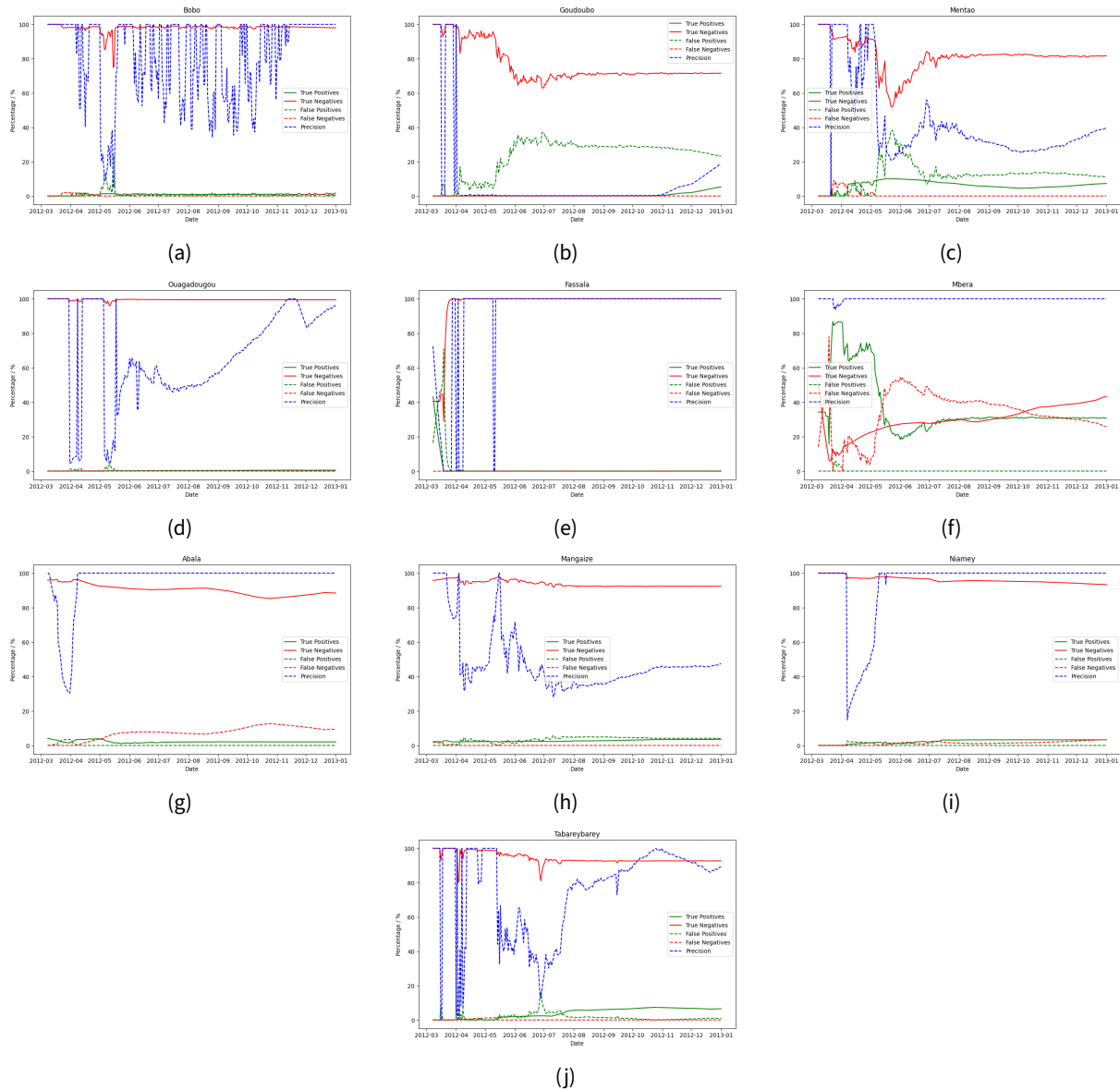


Figure 22: Classification and precision of GRASP results on camp populations for all camps.

As shown in Fig. 23, GRASP's classification precision ranges from 42% to 82%. This unweighted average does not account for higher precision in larger camps and their impact on overall model accuracy. Note that this graph is not identical to Fig. 8.

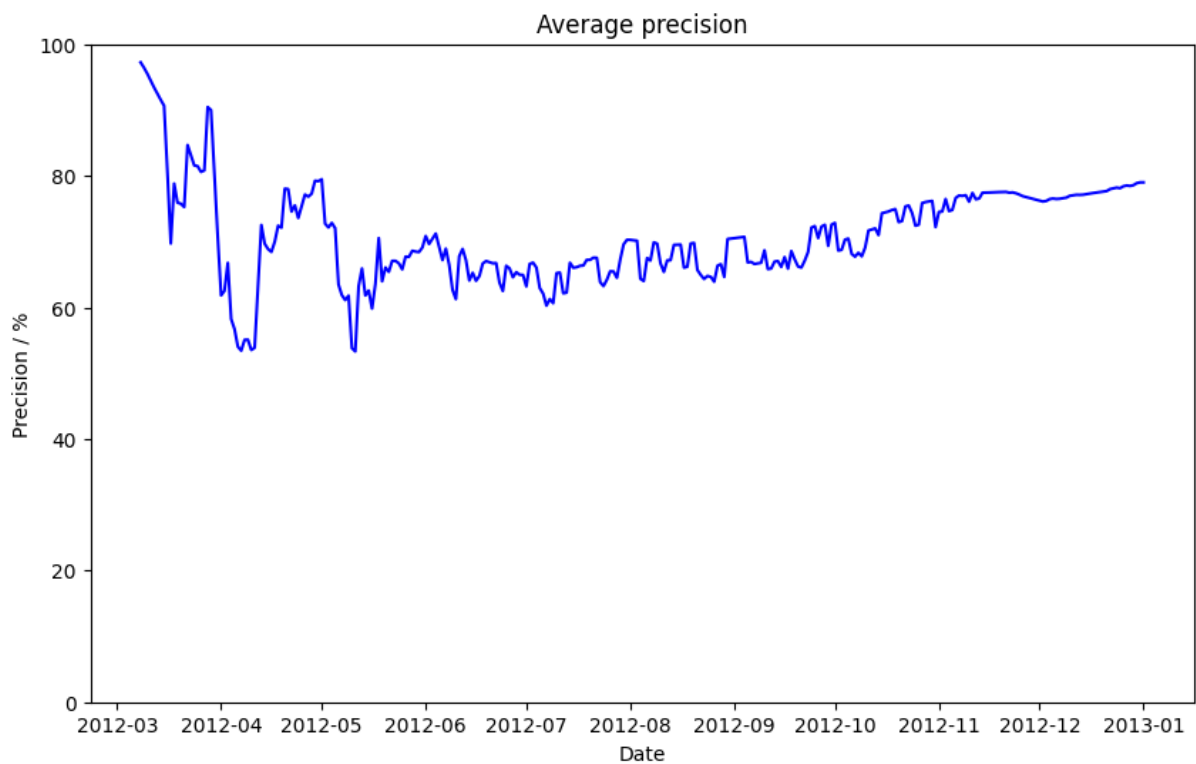


Figure 23: Average precision of GRASP results on camp populations for each camp

● Appendix G: Feature comparison for all camps

Fig. 24 compares the performance of GRASP model variants in predicting camp population distributions against FLEE and UNHCR data. Without the roulette method for destination selection, Bobo was not chosen after mid-May, and Mentaou, Ouagadougou, and Niamey were never chosen. This helps confirm that forced migrants don't always choose the nearest destination (Moore & Shellman 2007).

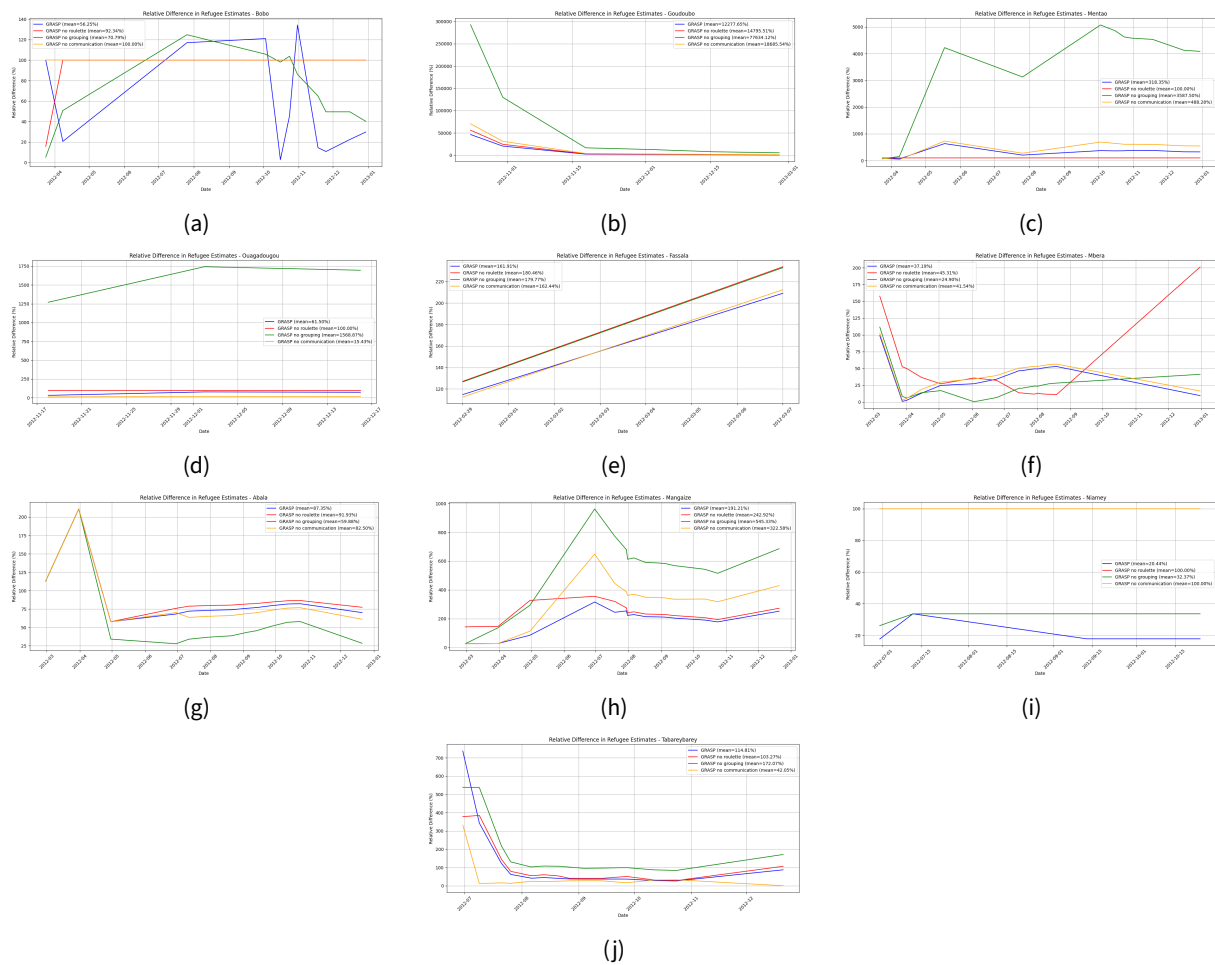


Figure 24: Comparison of GRASP versions, Flee, and UNHCR camp populations for all camps.

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